Bridging the Digital Divide: How 3G Internet Coverage Transforms Fertility Decisions in Nigeria

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Abstract

This paper studies how mobile internet access affects fertility and women's human capital in Nigeria, a high-fertility setting. Using the 2018 Nigerian Demographic and Health Survey, we reconstruct complete birth histories to follow women over the 2013–2018 period. These individual-level fertility trajectories are linked to georeferenced 3G coverage data from the GSMA, capturing one-year lagged measures of the staggered rollout of mobile broadband infrastructure across locations between 2012 and 2017. Our baseline estimates rely on a two-way fixed effects design, complemented by the Callaway-Sant'Anna heterogeneity-robust staggered difference-in-differences estimator. We find that a one standard deviation increase in local 3G coverage reduces the annual probability of birth among women aged 12–20 by 1.4–1.9 percentage points. These fertility declines operate primarily through delayed first cohabitation and delayed first birth, rather than increased contraceptive use. Consistent with a framework in which digital technology raises the opportunity cost of early childbearing, mobile internet exposure significantly improves educational attainment. Overall, the paper provides an integrated test of how information technology reshapes women's intertemporal allocation of time and highlights a demand-side economic opportunity channel through which digital infrastructure accelerates demographic transition in the Global South.

Keywords: Mobile Internet, Adolescent Fertility, Education

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1 Introduction

Contraceptive access programs have not closed Sub-Saharan Africa's fertility gap. Despite large-scale investments in family planning services, experimental evidence finds negligible effects of contraceptive provision on completed fertility in the region (Ashraf, Field and Lee, 2014; Dupas et al., 2024), suggesting that access is not the binding constraint on women's reproductive decisions. A complementary literature points to a deeper bottleneck, that information gaps can distort perceived opportunity costs of early childbearing (Dupas et al., 2024; Ashraf et al., 2021). In settings where women lack visible exposure to education pathways and alternative life trajectories, early family formation remains individually optimal even when contraceptives are available, because the alternative trajectory is neither observable nor salient (Bursztyn, Egorov and Jensen, 2019; Palivos, 2001; Becker, 1960; Doepke, 2004). Identifying what relaxes this information constraint, and through what channel, is the central question this paper addresses.

Mobile broadband expansion is a natural candidate for relaxing these information constraints, but the mechanism through which it might do so is ambiguous. Digital connectivity could operate through a supply-side channel, raising awareness of family planning options and increasing contraceptive adoption (Rotondi et al., 2020). Alternatively, it could act through a demand-side channel, making the returns to education more visible and raising the perceived opportunity cost of early family formation, which would delay first cohabitation and first birth without any change in contraceptive behavior (Becker, 1960; Doepke, 2004). Separating these channels requires data tracking fertility timing, contraceptive behavior, and human capital outcomes for the same population, a combination existing evidence from sub-Saharan Africa does not provide (Billari, Rotondi and Trinitapoli, 2020; Wildeman, Schrijner and Smits, 2023). We address this gap in the literature using quasi-random variation in 3G infrastructure across Nigeria, observing all three outcome margins for the same women.

Nigeria is well suited to this design: as Africa's most populous country, it offers high baseline fertility, a 3G rollout that expanded rapidly and unevenly across 36 states between 2013 and 2018, and a nationally representative survey with GPS-matched cluster locations. We use the 2018 Nigerian Demographic and Health Survey (NDHS), which covers 41,623 women of reproductive age and provides complete retrospective birth histories that allow us to reconstruct individual fertility trajectories from 2013 to 2018. These trajectories are linked to annual cell-tower-level 3G coverage maps from the GSMA Mobile Coverage database. The NDHS provides detailed information on education and contraceptive use alongside birth histories, i.e., the data combination required to track the key outcome margins for the same women. We measure 3G access as the average coverage share within a 20-kilometer buffer surrounding each survey cluster, computed annually against the GSMA maps.

Our empirical strategy uses a staggered difference-in-differences design. The identifying assumption is conditional parallel trends: absent 3G expansion, women in clusters receiving coverage at different times within the same state would have followed similar fertility trajectories, conditioning on climate controls and state-by-year fixed effects. Identification rests on within-state variation in rollout timing across clusters. We assess parallel trends empirically using pre-treatment placebo estimates and a power analysis following Roth et al. (2023), and find evidence supporting the parallel trends assumption. Because staggered adoption with heterogeneous treatment effects can lead two-way fixed effects estimation (TWFE) to place negative implicit weights on some cohort-specific treatment effects (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021), we complement TWFE with the Callaway and Sant'Anna (2021) estimator, which targets cohort-specific average treatment effects on the treated using never-treated clusters as controls and is consistent under arbitrary treatment effect heterogeneity.

Our results suggest that digital connectivity fundamentally transforms life-course trajectories among young women by delaying family formation and enabling substantial reallocation of time toward human capital accumulation. A one standard deviation increase in local 3G coverage reduces annual birth probability among women aged 12–20 by 1.4–1.9 percentage points, representing a 12.3–16.7 percent decline relative to baseline fertility rates. Placebo tests confirm that post-2013 3G coverage has no effect on pre-2013 birth rates, supporting the parallel trends assumption underlying our empirical approach. Critically, these fertility declines operate entirely through behavioral timing margins—delayed first cohabitation and postponed initial childbearing—rather than expanded use of reproductive technologies, as we find no significant effects of mobile internet exposure on adolescent contraceptive adoption.

The delayed family formation enables time reallocation that generates substantial human capital gains. Educational outcomes improve markedly among school-age cohorts exposed to mobile internet: primary completion increases by 7–10 percentage points across exposed cohorts, secondary completion rises by 12–16 percentage points, and the gap behind the official 12-year completion benchmark narrows by 1.1 years. These gains are concentrated among cohorts who were adolescents when 3G networks rolled out—precisely the group exhibiting the strongest fertility responses.

In addition, estimated fertility responses exhibit important heterogeneity along two dimensions. By reproductive history, mobile internet substantially delays entry into motherhood among nulliparous (i.e., yet to give birth) women while increasing subsequent fertility among young women who had already begun childbearing prior to network rollout. This pattern indicates that information technologies operate primarily as life-course timing devices rather than preference-altering interventions, with maximal effectiveness occurring before reproductive trajectories become established. By region, fertility reductions among adolescents are approximately twice as large in northern Nigeria as in the south, consistent with the north's higher baseline fertility, lower female education, and stronger barriers to alternative life trajectories. The north-south divergence reverses among women aged 20–25, reflecting compositional differences in partnership and childbearing status across regions.

Overall, these results demonstrate that telecommunications infrastructure generates demographic change through economic opportunity channels—primarily education—rather than contraceptive access pathways, and that the returns to connectivity are largest precisely where information barriers are highest.

Our analysis makes three fundamental contributions to the literature on telecommunications infrastructure and demographic outcomes in developing countries. First, we address endogeneity concerns that have limited previous research by exploiting plausibly exogenous variation in 3G coverage quality rather than self-selected adoption patterns. Existing studies rely on mobile phone ownership (Billari, Rotondi and Trinitapoli, 2020), self-reported internet usage (Wildeman, Schrijner and Smits, 2023), or adoption measures that are correlated with unobserved fertility preferences and household characteristics. We leverage technical features of telecommunications infrastructure that generate quasi-random variation in connection quality, following methodological approaches in Hjort and Poulsen (2019) who exploit submarine cable connections and Bahia et al. (2024) who use precise mobile broadband coverage data. This identification strategy builds on infrastructure-based approaches demonstrated in Kusumawardhani et al. (2023) for Indonesia, providing the first causally identified estimates of mobile internet's fertility effects in developing economies.

Second, we contribute to the literature on fertility and digital technology by identifying a distinct mechanism through which mobile internet reduces fertility in low-income settings. Whereas much of the family planning literature emphasizes supply-side interventions and contraceptive access, we show that fertility declines primarily through delayed first cohabitation and first birth rather than increased contraceptive use. This mechanism reflects fundamental constraints on women's reproductive autonomy in the Global South, where women often lack decision-making power over contraceptive adoption during sexual encounters. By exploiting detailed birth histories, we directly test this implication of our conceptual framework and show that mobile internet significantly delays fertility timing without affecting contraceptive behavior. This finding contrasts with evidence from developed-country contexts, where fertility responses operate through direct information effects Guldi and Herbst (2017) or work flexibility that facilitates higher fertility Billari, Giuntella and Stella (2019). Instead, our results align with emerging evidence from Sub-Saharan Africa documenting technology-induced changes in fertility timing and preferences (Billari, Rotondi and Trinitapoli, 2020; Wildeman, Schrijner and Smits, 2023). Relative to studies focusing on health knowledge or contraceptive uptake (Rotondi et al., 2020; Pesando, 2022), we highlight an alternative demand-side pathway—economic empowerment—that is particularly relevant in contexts with restrictive gender norms.

Third, we contribute to the literature by providing the first integrated test of a conceptual framework in which technology reshapes women's fertility and education decisions through intertemporal time allocation, with direct implications for economic development in the Global South. Existing work on innovation-driven growth emphasizes the role of knowledge access and creative destruction in raising productivity (Mokyr, 2018; Aghion and Howitt, 1992; Howitt, 1999), while related microeconomic studies examine technology's effects on schooling Jensen (2010) or fertility preferences in isolation. In contrast, our paper empirically tests how mobile internet simultaneously alters women's decisions across fertility timing and educational investment, showing that these margins are tightly linked through opportunity-cost channels. We find that technology exposure significantly improves schooling progression and delays fertility—effects concentrated during adolescence, when educational and reproductive trajectories are most malleable. Compared to prior studies that focus on fertility outcomes alone (Guldi and Herbst, 2017), our results demonstrate that digital infrastructure empowers women to reallocate time away from early childbearing toward human capital accumulation, with education emerging as the primary channel through which demographic transition accelerates.

The remainder of the paper proceeds as follows. Section 2 provides background on Nigeria's fertility context and the expansion of telecommunications infrastructure, and introduces the conceptual framework. Section 3 describes the data sources, and Section 4 outlines the empirical strategy. Section 5 presents the main results on fertility outcomes and robustness checks. Section 6 and Section 7 investigate the underlying mechanisms, focusing on marriage and birth timing, women's education, and contraceptive adoption. Section 8 examines heterogeneity by reproductive history and region. Finally, Section 9 concludes with policy implications and directions for future research.

2 Background and Conceptual Framework

High Fertility Rates and Reproductive Constraints in Nigeria

Nigeria has one of the highest fertility rates globally, at 4.5 children per woman as of 2023 (United Nations Population Division, 2024), and remains among the last countries to begin the demographic transition. As Africa's most populous nation with an estimated 237.5 million people in 2025 (United Nations Population Fund, 2025), these fertility patterns reflect a context in which social, economic, and institutional factors shape women's reproductive and life-course decisions. Modern contraceptive prevalence stands at 12% among married women of reproductive age, below the sub-Saharan African average of 28% (Secretariat, 2023). While supply-side access is an issue, it does not account for this gap alone. Social norms in northern regions vest reproductive decision-making primarily in male partners and extended family members, creating barriers to contraceptive adoption that go beyond availability. These norms intersect with low formal economic participation: women's formal employment rate stands at 22.7%, compared to 64.1% for men (International Labour Organization, 2023)—most women work in informal or unpaid activities, limiting scope for independent economic decision-making within households.

Early marriage is widespread in Nigeria. 43% of Nigerian women are married before age 18, with over 60% in northern states (United Nations Children's Fund, 2022). Early marriage is associated with lower school enrollment, as only 4% of married girls aged 15–19 remain in school, compared to 69% of unmarried girls (National Population Commission, 2019). Early marriage constrains access to formal employment and the economic independence relevant to fertility timing decisions. These patterns are most pronounced in northern Nigeria, where practices such as female seclusion (purdah) limit women's mobility and access to information about opportunities outside the household (Bloom, Kuhn and Prettner, 2017). Regional differences in fertility reflect these structural disparities: northern states report total fertility rates above 6 children per woman, female literacy below 30%, and modern contraceptive prevalence below 5%, while southern states show lower fertility rates (3–4 children per woman) and higher education and employment levels (National Population Commission, 2019). This variation indicates that information constraints and limited economic opportunities—rather than contraceptive supply alone—are central to understanding Nigeria's fertility patterns.

Nigeria's high fertility rates coincide with high poverty and unemployment prevalence. Over 40% of the population lives below the extreme poverty line (The World Bank, 2022), and youth unemployment exceeds 40% (National Bureau of Statistics, 2023). Nigeria ranks 128th out of 146 countries in the Global Gender Gap Index (World Economic Forum, 2023), a ranking that reflects both economic and social dimensions of the gender gap. These patterns suggest that expanding women's economic opportunities may be central to fertility change, independent of formal family planning programs.

The Evolution and Expansion of Mobile Telecommunications Infrastructure in Nigeria

Nigeria's telecommunications sector expanded rapidly following market liberalization in 2001. Prior to liberalization, the sector was underdeveloped, with fewer than 500,000 fixed telephone lines serving a population of over 120 million in 2000—a teledensity below 0.4

lines per 100 inhabitants (International Telecommunication Union , ITU). The entry of private mobile operators through competitive licensing restructured this landscape: active mobile subscriptions reached approximately 198 million by 2020 (Nigerian Communications Commission, 2020). Mobile technology in Nigeria developed in successive generations, each enabling more sophisticated services. The first wave, beginning in 2001–2002, introduced 2G (GSM) networks providing basic voice and SMS capabilities (Nigerian Communications Commission, 2021). 2G offered limited data capabilities, with connection speeds typically below 64 kilobits per second, and could not support the kind of internet use relevant to this study.

3G mobile internet represented a qualitative improvement over 2G. Based on UMTS and HSPA technologies, 3G services began commercial deployment in major urban centers around 2008, initially in Lagos, Abuja, and Port Harcourt (Nigerian Communications Commission, 2020). 3G supports data speeds of 384 kilobits per second to several megabits per second, enabling internet browsing, video streaming, and social media use that 2G could not support (Aker and Ksoll, 2016; Jensen, 2012). This connectivity provides access to information and social networks beyond traditional community boundaries—including information about employment opportunities, educational resources, and peer networks (Jensen, 2012; Aker and Ksoll, 2016). For women, 3G mobile internet can provide access to such information independently of household or community gatekeepers. Data costs declined over the study period, from over ₦1,000 (approximately $3) per gigabyte in 2010 to under ₦200 ($0.50) by 2020, reflecting increased competition and infrastructure investment (Web Foundation, 2021).

By 2013, 3G coverage reached approximately 65% of Nigeria's population, rising to 85% by 2018 and over 90% by 2020 (Nigerian Communications Commission, 2020). Regional disparities, however, remained substantial. Operators—MTN, Airtel, Globacom, and 9mobile—prioritized commercially viable markets, beginning with high-density urban areas

(Bahia et al., 2024). Lagos State achieved broad coverage by 2010, followed by other south-western states and the Federal Capital Territory; many rural locations in northern states did not receive 3G access until after 2015 (Nigerian Communications Commission, 2020). This staged rollout reflects technical and economic constraints: 3G infrastructure requires more capital-intensive base station equipment than 2G networks, and the fiber optic back-haul connections needed to support 3G data traffic added cost in areas with limited existing infrastructure (Aker and Ksoll, 2016). The result was variation in coverage timing across clusters that correlates with population density, economic development, and prior infrastructure—variation that our identification strategy exploits.

Conceptual Framework

Women's fertility, education, and employment decisions are jointly determined by labor-market opportunities. In many developing countries, female employment is concentrated in unpaid family work and informal self-employment (Banerjee and Duflo, 2007; Fields, 2011). Limited access to wage and skill-intensive jobs lowers the returns to schooling and the opportunity cost of early family formation (Goldin, 1994; Doepke and Tertilt, 2018). Mobile internet may alter these incentives by improving access to information and formal employment opportunities.

Consider a woman who first chooses schooling and marriage timing and then, conditional on marriage, chooses fertility. Early marriage may restrict continued schooling and subsequent employment. Her per-period utility is

$$U_a = u(C_a) - \phi_s s_a - \phi_b b_a,$$

where s_a denotes schooling enrollment and b_a indicates childbearing. The parameter ϕ_b represents the net opportunity cost of a birth—the direct and forgone-earnings costs of childbearing net of its utility benefit. When formal employment opportunities are limited, this net cost may be negative, $\phi_{b0} < 0$, making early fertility optimal.

Human capital evolves according to

$$h_{a+1} = h_a + \eta s_a,$$

where η is the productivity of schooling. Mobile internet does not necessarily increase η directly. Instead, it may raise perceived returns to education by expanding access to information about schooling and employment opportunities, thereby increasing optimal enrollment (Jensen, 2010).

Most women have access to an informal or unpaid employment option yielding w_0. Formal or moderate-skill employment provides $w_1(h_a) > w_0$, with $w_1'(h_a) > 0$. Let $\pi(T_c)$ denote the probability of obtaining such employment, where T_c is local mobile internet access and $\pi'(T_c) > 0$. Expected earnings are

$$\mathbb{E}[w(T_c, h_a)] = \pi(T_c)w_1(h_a) + [1 - \pi(T_c)]w_0.$$

Mobile internet therefore primarily changes employment composition by shifting women from informal or unpaid work toward wage and moderate-skill jobs, rather than necessarily increasing overall employment.

The opportunity cost of early childbearing is

$$\phi_b(T_c, h_a) = \phi_{b0} + \lambda\pi(T_c)\big[w_1(h_a) - w_0\big],$$

where $\lambda > 0$. A woman delays childbearing when $\phi_b(T_c, h_a) > 0$. The corresponding coverage threshold satisfies

$$\pi(\widehat{T}_c) = \frac{|\phi_{b0}|}{\lambda\big[w_1(h_a) - w_0\big]}.$$

Thus, greater mobile internet access increases the opportunity cost of early family formation by improving access to higher-return employment. The mechanism requires women to have sufficient agency to delay marriage but does not require unilateral control over fertility within marriage.

The framework generates three main predictions. First, mobile internet delays marriage and fertility, particularly among younger women whose schooling and family-formation decisions remain adjustable. Second, it shifts employment from informal or unpaid work toward wage and moderate-skill employment, with smaller effects on overall employment. Third, because the primary mechanism operates through delayed marriage, fertility may decline without a corresponding increase in contraception among women who are already married or cohabiting.

3 Data

Nigerian Demographic and Health Surveys Program (NDHS)

Our main study sample is comprised of 41,623 women from the 2018 wave of the Nigerian Demographic and Health Surveys (NDHS), which coincides with the period of Nigeria's 3G network expansion.[1] The NDHS uses stratified two-stage cluster sampling for national representativeness, selecting enumeration areas proportional to size, then randomly sampling 25–30 households per area. The survey provides comprehensive nationwide coverage across all 36 states and the Federal Capital Territory, with denser sampling in populous southern regions and comparatively fewer sampling points in northern states reflecting lower population density. The 2018 survey included 41,623 women of reproductive age (15–49 years), providing high-quality data on fertility, education, wealth, and mobile phone access that we match with geospatial mobile coverage data. We use the GPS coordinates provided by the NDHS for each survey cluster, which include a random offset (up to 2km for urban clusters

[1] See National Population Commission (2019) for the wave-level sample description. The 2018 NDHS interviewed 41,623 women of reproductive age (15–49 years). We use the 2018 wave because it provides the most complete retrospective birth histories over the 2013–2018 study period.

and 5km for rural clusters) to ensure respondent confidentiality.[2]

Women's Fertility Our primary outcome variable, women's fertility, is derived from the individual-level birth history panels in Figure A1. These panels, a standard feature of the DHS, provide a retrospective record of all live births for each surveyed woman. We use detailed birth history information to construct an annual woman–year panel, indicating whether each woman gives birth in each year from 2013 through 2018. From this panel, we generate our main outcome: an annual dummy variable that takes a value of 1 if a woman gave birth in a given year, and 0 otherwise. Women with no recorded births are assigned values of 0 for all years. We also construct binary indicators for the timing of first cohabitation and first birth to analyze behavioral margins.

Figure 1 shows the spatial-temporal patterns of birth rates across Nigeria from 2013 to 2018. Birth rates exhibit a persistent north-south gradient, with northern regions consistently showing higher fertility compared to Southern areas. This geographic pattern remains relatively stable throughout the observation period, along with a modest overall decline in birth rates over time, particularly visible in some northern regions.

[2]The DHS displaces urban cluster coordinates by up to 2km and rural clusters by up to 5km to protect respondent privacy. This random displacement introduces a form of measurement error that is orthogonal to our variables of interest and thus should not systematically bias our regression estimates. Recent methodological studies confirm that these privacy protection techniques do not significantly impact regression estimates on average (Michler et al., 2022).

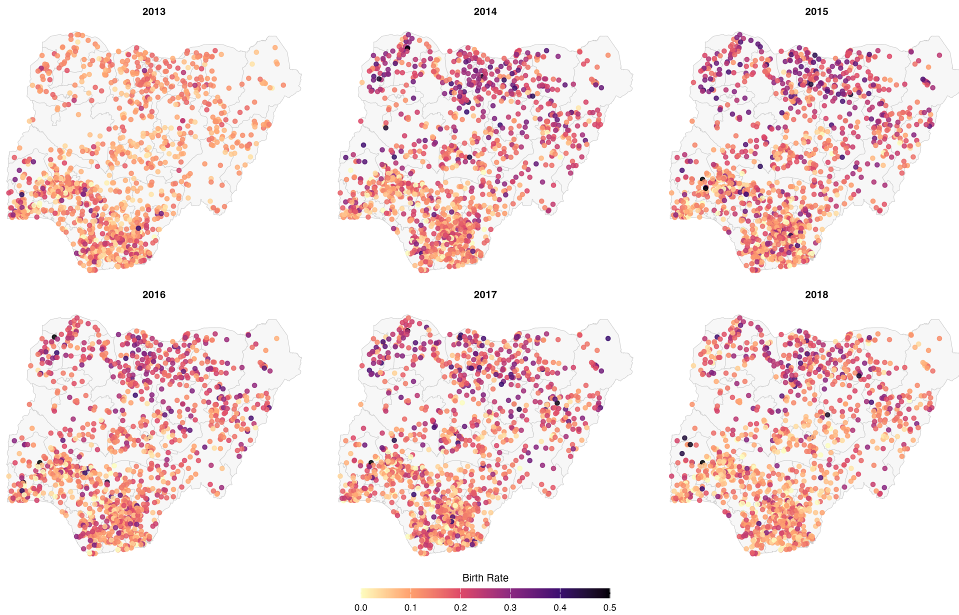

Figure 1: Heatmap of Birth Rate Across Clusters (2013–2018).

Reproductive Behaviors To test the family planning channel, we construct variables for reproductive behavior. The NDHS asks women about their use of family planning methods. We create indicators for the use of *any* method, *modern* methods (including pills, IUDs, injectables, implants, and condoms), and *traditional* methods (including periodic abstinence and withdrawal).

Women's Education We construct multiple education outcomes from DHS variables while carefully addressing data quality limitations inherent in the survey design. Using women's reported highest education level and years of schooling, we create binary indicators for key attainment milestones consistent with Nigeria's 6–3–3 education system: primary completion (grades 1–6), secondary completion (grades 7–12), and higher education. We additionally construct an education gap measure that captures delays in schooling progression, defined as the difference between expected years of education—given by the minimum of age minus six and twelve years—and actual years completed. This framework allows us to examine both extensive-margin effects (educational completion) and intensive-margin effects (grade progression) of digital infrastructure on human capital accumulation.

Socioeconomic Status Finally, we use a standard set of socioeconomic and demographic variables from the DHS as controls for the cross sectional analysis when panel data are not available. These include the woman's age, regions, religion and a household wealth index constructed by the DHS from data on asset ownership and housing characteristics. Table 1 summarizes baseline demographic, socioeconomic, and empowerment characteristics of women in the 2018 DHS sample. Women in the sample are on average 29 years old, with educational attainment spanning from no formal education (34.5%) to higher education (10.4%). About 37% of respondents reside in northern regions and 50% are Muslim. Fertility patterns indicate an average of 1.3 children under age five and roughly three total children per woman, with first births occurring around age 20 and first marriages at age 19. Labor market participation is relatively high, with 65% of women currently employed.

3G Mobile Internet Coverage

3G mobile internet coverage is processed by integrating Demographic and Health Survey (DHS) locations with GSM (Global System for Mobile Communications) coverage data in Nigeria. We use 3G coverage data released annually, where each year's release (e.g., 2010) contains operator data collected through the end of the previous year (2009). Using DHS cluster points from the 2018 survey, the methodology creates multiple circular buffer zones at varying distances (0.5 to 100 kilometers) around each survey cluster.[3] These locations are first transformed into the UTM 32 projection system, appropriate for Nigeria's geographic position. For each buffer zone, the code processes mobile coverage data from GSM TIF files, calculating coverage values through a spatial analysis that counts non-NA cells within each buffer and normalizes these values relative to the maximum count. The analysis generates

[3] Coverage is assigned to each woman based on her survey-date cluster location. To the extent that women migrated toward covered areas over the study period — particularly if migration was driven by economic opportunity correlated with 3G expansion — this assignment may overstate historical coverage for earlier years in the birth history. We address this concern in robustness checks by restricting the sample to women who report residing in the same location for five or more years (Section 5).

a comprehensive dataset recording the coverage (GSMCOVER), year (GSMYEAR), generation type (GSMGEN), and buffer distance (BUFFERDIST) for each DHS cluster point.

Figure 2 illustrates the expansion of 3G mobile network coverage across Nigeria between 2012 and 2018, providing the key variation for our identification strategy. Each dot represents a survey cluster from the NDHS, with colors indicating the proportion of 3G coverage within a 20-kilometer radius of each cluster. In 2012, coverage exhibited a clear north-south gradient, with minimal coverage concentrated in northern areas and higher coverage clustered in southern regions. By 2018, this pattern persisted but with substantial overall expansion—southern areas developed contiguous zones of high connectivity, while northern regions experienced considerable improvement from minimal to low-moderate coverage levels. This pattern of expansion was not uniform across all regions, creating variation that our empirical analysis exploits. The heterogeneous rollout of 3G networks—with some areas gaining access early, others later, and some remaining with limited coverage—allows us to examine how differential exposure to mobile internet affects fertility decisions while controlling for time-invariant regional characteristics through our two-way fixed effects models.

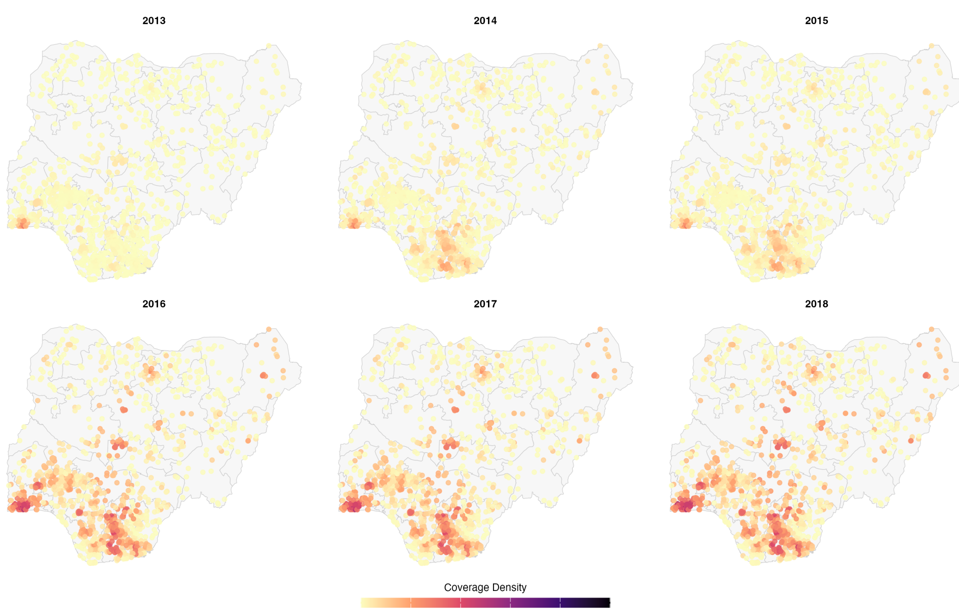

Figure 2: Heatmap of 3G Coverage Across 20-Kilometer Radius of Clusters (2012–2018).

Climate and Weather Controls

We incorporate comprehensive climate and weather data from Copernicus's ERA5 reanalysis product to control for environmental factors that may influence both telecommunications infrastructure deployment and economic activity patterns.[4] Daily data on 2-meter air temperature, precipitation flux, 10-meter wind speed, vapor pressure, solar radiation, and rainfall duration are collected and processed to generate cluster-level climate measures. The raw daily observations are first aggregated to monthly values, then summarized annually by calculating means for temperature, wind speed, vapor pressure, solar radiation, apparent temperature, and wet bulb temperature, while precipitation variables are summed across months. These controls help isolate the effects of mobile internet coverage from underlying environmental factors that could affect both infrastructure rollout decisions and demographic or economic outcomes.

4 Empirical Strategy

We estimate the effect of 3G mobile internet access on annual birth probability, exploiting variation in the timing of coverage rollout across Nigerian clusters between 2013 and 2018. The treatment variable is the standardized share of 3G coverage within a 20-kilometer buffer of each cluster, which varies across clusters and over time as commercial operators expanded their networks. Observational comparisons of coverage and fertility could potentially be confounded by endogenous infrastructure placement: operators target commercially attractive areas that may differ systematically in their demographic trajectories, and unobserved economic development simultaneously drives both coverage expansion and reproductive behavior. We address this concern using a staggered difference-in-differences design that relies on conditional parallel trends as the identifying assumption. The within-state comparison

[4] ERA5 is the latest climate reanalysis produced by ECMWF, providing hourly data on many atmospheric, land-surface and sea-state parameters together with estimates of uncertainty. The dataset is available at: https://climate.copernicus.eu/climate-reanalysis.

is central to that argument: rollout pace across clusters within a state reflected logistical and topographical constraints more than local demand conditions, providing design-based motivation for why treated and never-treated clusters would have followed parallel fertility trends absent coverage expansion. We test this assumption empirically rather than relying on the design argument alone, and implement the design using three estimators described below.

The analysis includes women of reproductive age (12–49 years). Since a birth recorded in year t reflects a conception decision made roughly nine months earlier, we align the treatment variable with the period of conception by using one-year lagged coverage. The estimating equation is:

$$Birth_{ict} = \alpha_i + \delta_{st} + \beta \times Coverage_{ct-1} + \mathbf{W}'_{ct-1} \times \gamma + \varepsilon_{ict} \tag{1}$$

where $Birth_{ict}$ is a binary indicator equal to one if woman i in cluster c gives birth in year t, and $Coverage_{ct-1}$ is the standardized 3G coverage share within a 20-kilometer buffer of cluster c in the preceding year. The vector $\mathbf{W}_{ct-1}$ includes time-varying climate controls: precipitation, solar radiation, wind velocity, vapor pressure, temperature, and rainfall duration. Individual fixed effects α_i absorb all time-invariant heterogeneity, including permanent differences in socioeconomic status, preferences, and fertility. State-by-year fixed effects δ_{st} absorb all shocks common to clusters within the same state in a given year, including state-level variation in economic conditions, public health programs, and governance. The coefficient β is identified from within-individual changes in cluster-level coverage, conditional on these controls.

The key identification assumption is conditional parallel trends: in the absence of 3G expansion, women in clusters with different coverage trajectories would have experienced similar fertility trends, conditional on individual fixed effects α_i, state-by-year fixed effects δ_{st}, and time-varying climate controls $\mathbf{W}_{ct-1}$. Individual fixed effects absorb permanent between-cluster differences in preferences, fertility, and socioeconomic status; state-by-year fixed effects absorb all state-level time-varying shocks. Identification then requires only that within-state rollout timing be uncorrelated with cluster-level fertility shocks not captured by these controls. Three threats motivate robustness checks reported in Section 5: within-state targeting of economically accelerating clusters, migration endogeneity where a woman's survey cluster may differ from her cluster during the treatment window, and spatial spillovers from overlapping 20-kilometer coverage buffers. We assess the plausibility of this assumption through pre-treatment placebo estimates and a power analysis following Roth et al. (2023); Section 5 reports these tests.

We use two primary estimators to estimate the effects, both grounded in the staggered rollout design of mobile internet expansion and relying on the conditional parallel trends assumption. Standard two-way fixed effects estimation of equation (1) is consistent for β under conditional parallel trends and treatment effect homogeneity; under heterogeneous treatment effects and staggered adoption, however, OLS can place negative implicit weights on some cohort-specific effects, potentially attenuating or reversing the sign of the aggregate estimate (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Borusyak, Jaravel and Spiess, 2024). The TWFE estimator therefore serves as a benchmark whose continuous dose-response parameterization is most directly comparable to prior literature. The Callaway-Sant'Anna estimator avoids negative implicit weights by construction; convergence between the two constitutes corroborating evidence that heterogeneity bias does not materially contaminate the benchmark estimate.

To provide a benchmark, we estimate equation (1) using TWFE, clustering standard errors at the cluster level. The preferred specification replaces year fixed effects δ_t with state-by-year interactions δ_{st}; we also report specifications with year fixed effects and birth cohort fixed effects. Under conditional parallel trends and homogeneous treatment effects across adoption cohorts, TWFE identifies the ATT: the causal effect of a unit increase in standardized 3G coverage share on annual birth probability. The continuous treatment parameterization is the natural specification given that coverage varies continuously across clusters and time, and it yields a dose-response estimand that is directly comparable to prior studies of mobile internet and demographic outcomes.

We give priority to the Callaway-Sant'Anna (Callaway and Sant'Anna, 2021) estimator as our headline binary-treatment estimate because it avoids negative implicit weights by restricting all comparisons to never-treated clusters, directly targets a cohort-specific treatment effect, and remains consistent under arbitrary treatment effect heterogeneity across the rollout timeline—properties that TWFE does not guarantee under staggered adoption. Let G_i denote the first period in which individual i's cluster receives positive 3G coverage, and let $G_i = \infty$ denote clusters never treated during the sample period. The group-time average treatment effect is:

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid G_i = g]\,, \tag{2}$$

where untreated potential outcomes are estimated using never-treated units. Identification requires:

$$\mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid G_i = g] = \mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid G_i = \infty], \tag{3}$$

conditional on observed covariates. Treatment is defined at the cluster level based on first exposure to nonzero 3G coverage within the 20-kilometer buffer. Using never-treated clusters as the sole comparison group avoids the contamination that arises when already-treated units serve as implicit controls for later-treated cohorts (Callaway and Sant'Anna, 2021). Relaxing cohort homogeneity allows the estimand to reflect genuine variation in treatment effects across the rollout timeline. We aggregate group-time $\mathrm{ATT}(g,t)$ estimates using the simple-average aggregation scheme of Callaway and Sant'Anna (2021) and report the resulting point estimate as our summary measure of the binary treatment effect. We assess the plausibility of parallel trends by estimating placebo $\mathrm{ATT}(g,t)$ for all $t < g$ and testing whether the pre-treatment estimates are jointly indistinguishable from zero. A power analysis following Roth et al. (2023) bounds the magnitude of pre-trends detectable with

50% power. Section 5 compares this bound to the estimated treatment effect magnitude to assess whether undetected pre-trends could account for the findings. As an internal consistency check, we also estimate a binary TWFE specification that replaces $Coverage_{ct-1}$ with the indicator $\mathbf{1}[Coverage_{ct-1} > 0]$, equaling one whenever a cluster has any 3G coverage in the preceding year. This specification uses the same treatment parameterization as the Callaway-Sant'Anna estimator and provide a benchmark of the extent to which cohort-treatment heterogeneity contaminates the TWFE comparison group.

Additional robustness checks are reported in Section 5 and the appendix. These include alternative spatial buffer definitions, a nighttime light density control, a 2G network falsification test, and a sample restricted to long-term residents. We further validate the main estimates using Double Machine Learning (DML) with four machine-learning methods (LASSO, ElasticNet, Random Forest, and XGBoost) under a partially linear regression specification, following Chernozhukov et al. (2018). The DML approach removes potential confounding through flexible first-stage prediction of both the outcome and the treatment, providing an alternative nonparametric benchmark that does not impose the linearity constraints of TWFE.

5 Fertility Effects of Mobile Internet

Results for Two-Way Fixed Effect Model

We begin by examining the average treatment effects of mobile internet coverage on fertility outcomes to establish the baseline patterns, while also analyzing effects across different age groups to capture potential heterogeneity in how women at various life stages respond to digital connectivity. Table 2 presents the impact of mobile internet coverage on fertility across age groups using a 20km buffer distance around survey clusters to measure 3G coverage exposure. The full sample results (Panel A) show small and mixed effects, with coefficients ranging from -0.001 to -0.005 percentage points, masking important age-based heterogeneity in responses to mobile internet access.

The most striking finding emerges among women aged 12–20 (Panel B), where mobile internet access demonstrates strong and consistent negative effects on fertility. A one standard deviation increase in 3G coverage significantly reduces birth probability by 1.4–1.9 percentage points ($p < 0.01$ across all specifications). Given a birth rate of 11.4% among areas without mobile coverage, these effects represent substantial reductions of 12.3–16.7% relative to the control group mean. The consistency of these effects across specifications with different fixed effects structures—including individual fixed effects, state-by-year fixed effects, and age cohort fixed effects—provides strong evidence for the robustness of this relationship.

In contrast, women aged 20-25 (Panel C) show no significant response to mobile internet coverage across all specifications, despite having the highest baseline fertility rate (28.9%). The coefficients are small in magnitude and statistically insignificant, ranging from -0.006 to 0.007 percentage points. This null result suggests that women in peak reproductive years have already formed fertility preferences that are less susceptible to information-based interventions, or that the opportunity costs of childbearing may be lower for this group who have likely completed their education and are establishing careers. For women over 25 (Panel D), results indicate small positive but statistically insignificant effects across all specifications. This suggests that mobile internet access may help older women overcome information constraints to achieve desired fertility levels, though the evidence is not robust enough to draw strong conclusions.

The analysis reveals that mobile internet's impact on fertility is highly age-dependent, with the strongest effects concentrated among women aged 12-20, and we demonstrate robust results across alternative buffer distances as shown in Figure A2. These divergent patterns suggest that mobile internet operates as a transformative technology primarily during the critical transition from adolescence to early adulthood, when life course trajectories are most malleable and the economic returns to delaying childbearing are highest, while having minimal impact on women who have already established their reproductive preferences and family formation patterns.

Placebo Test To test for parallel trends, we conduct a placebo test using our two-way fixed effects (TWFE) specification. Since our main analysis covers the period of 3G network expansion in Nigeria (2013–2018), we test whether pre-period fertility outcomes from 2006–2011 are systematically related to these subsequent coverage patterns. Under the parallel trends assumption, we should find no relationship between future treatment and pre-treatment outcomes—that is, the null hypothesis of parallel pre-trends ($\beta = 0$) should hold. We apply our baseline specification from Equation (1) to historical birth outcomes from 2006–2011, regressing them on future 3G coverage from 2013–2018, thereby testing for pre-existing trends over a period that predates actual network deployment.

Table 2 presents these results across age groups and specifications. Across demographic subgroups and preferred specifications (columns 2–3), coefficients are small and statistically indistinguishable from zero. The full-sample coefficient in the baseline specification without state-by-year fixed effects (column 1, Panel E: -0.006^{***}) is significant, but becomes small and insignificant once state-by-year fixed effects are included, indicating that the column 1 estimate reflects confounding by state-level trends rather than a true pre-existing pattern. These results indicate that we cannot reject the null hypothesis of parallel pre-trends in our preferred specifications, consistent with the identifying assumption. The Age 12–20 placebo sample has limited power because fertility is rare at these ages during the 2006–2011 period. The absence of systematic pre-existing differential trends between areas with higher and lower future 3G coverage strengthens confidence in our identification strategy and suggests that our main estimates are unlikely to be confounded by time-varying unobservables correlated with network expansion trajectories and underlying fertility patterns.

Event Study Approach and Assessment of Pre-Trends

To accommodate treatment effect heterogeneity across time and treated units while testing for parallel trends, we employ the Callaway and Sant'Anna (2021) estimator, which is robust to heterogeneous treatment effects under staggered adoption. This method addresses concerns about bias in standard TWFE models when treatment effects vary across groups and time periods. We focus our causal interpretation on women aged 12–20.

Visual Evidence of Parallel Trends and Dynamic Treatment Effects Figure 3 presents event study coefficients for women aged 12–20 using the Callaway and Sant'Anna (2021) estimator. The plot provides visual evidence consistent with the parallel trends assumption. Pre-treatment coefficients spanning periods -4 to -2 cluster around zero with confidence intervals encompassing the null, exhibiting no discernible trend. Post-treatment coefficients reveal economically significant and persistent negative effects: modest at period 0, growing through period 1, and remaining negative through period 3. This dynamic pattern suggests that fertility responses to mobile internet access intensify with exposure duration, consistent with models where information acquisition and behavioral adaptation occur gradually.

Formal Statistical Tests Table 3 reports formal statistical tests validating the visual evidence. The joint pre-trend test fails to reject the parallel trends assumption for women aged 12–20, yielding $\chi^2(4) = 2.02$, $p = 0.731$. We implement Roth (2022) power analysis to assess whether null pre-trend findings reflect genuine parallel trends rather than insufficient statistical power. For the 12–20 age group, linear pre-trend slopes detectable with 50% power are 0.006, with Bayes factors of 0.564 favoring the null hypothesis of parallel trends over linear violations of this magnitude. Together with the insignificant joint pre-trend test, these results provide little evidence that differential pre-treatment trends drive the estimated fertility effects.

The Callaway-Sant'Anna average treatment effect is -0.021 (SE $= 0.008$, $p < 0.01$), indicating that 3G coverage expansion significantly reduces fertility among adolescent women. Effects are robust to alternative identifying assumptions and heterogeneous treatment effect concerns. Corresponding event study estimates for the full sample and additional age cohorts are provided in the appendix (Figure A3).

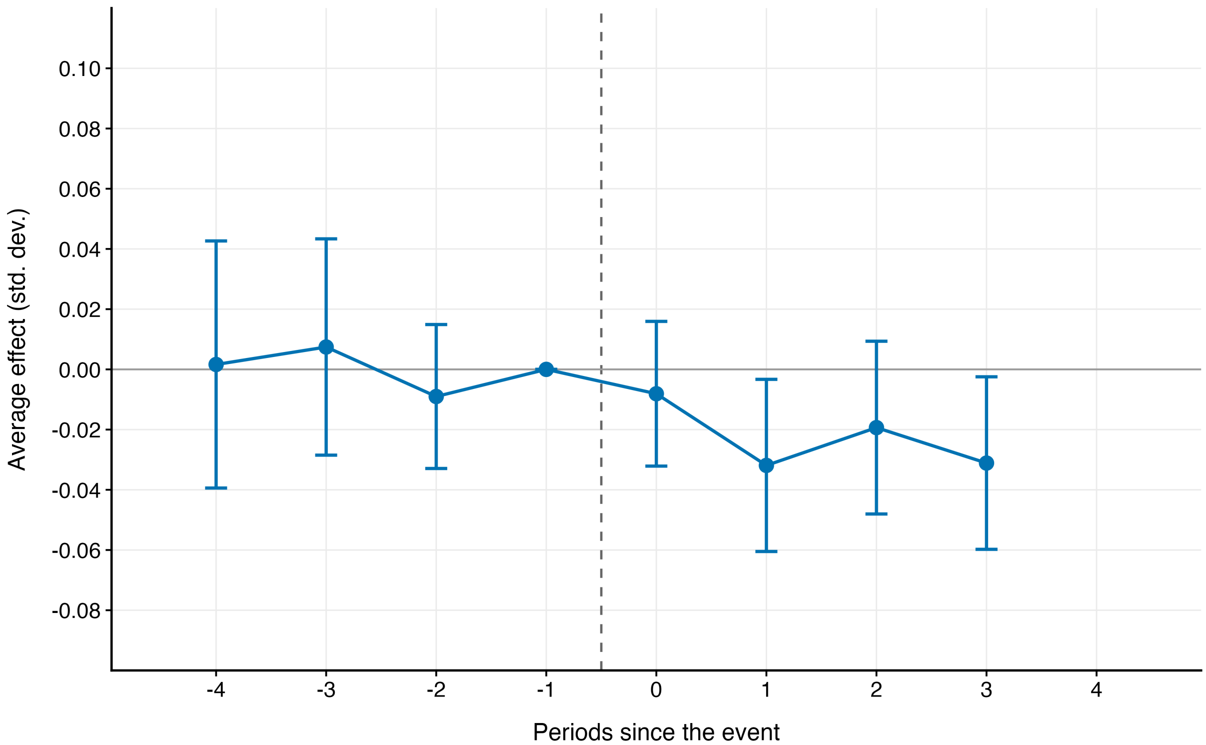

Figure 3: Event Study Estimates of 3G Coverage Effects on Fertility (Ages 12–20)

Notes: This figure presents event-study estimates for women aged 12–20 using the Callaway and Sant'Anna (2021) heterogeneity-robust difference-in-differences estimator. Period -1 serves as the reference period. Pre-treatment coefficients (periods -4 to -2) test the parallel trends assumption; post-treatment coefficients (periods 0 to +3) estimate dynamic treatment effects relative to first 3G coverage exposure within 20km. The estimator uses never-treated clusters as the comparison group and controls for 2012 baseline weather variables and age. The average treatment effect is ATT $= -0.021$ (SE $= 0.008$, $p < 0.01$) with joint pre-trend test p-value $= 0.731$. Error bars represent 95% confidence intervals with standard errors clustered at the cluster level.

Table A1 in the appendix reports Double Machine Learning robustness checks using four machine-learning methods (LASSO, ElasticNet, Random Forest, and XGBoost). Across all methods, the estimated effect for women aged 12–20 is negative and statistically significant ($p < 0.01$), ranging from -0.018 to -0.024, consistent with the TWFE and CS-DiD estimates.

Robustness Checks

We conduct several robustness exercises to assess the validity of our identification strategy and the sensitivity of our estimates to alternative specifications. These tests address potential concerns regarding measurement choices, confounding factors, and identification assumptions.

Alternative Buffer Distance Specifications

We examine the sensitivity of our results to the choice of geographic buffer distance used to measure 3G coverage exposure. Our baseline specification employs a 20km radius around survey clusters, but this choice may influence the magnitude and precision of our estimates. We re-estimate our main specification using buffer distances ranging from 5km to 50km to assess the robustness of our findings across different spatial definitions of treatment exposure. Figure A2 presents estimated coefficients and 95% confidence intervals across buffer distances for each age subgroup. The results exhibit considerable stability in magnitude and statistical significance across the examined range. For the full sample, coefficients remain small and consistently negative. Among women aged 12–20, negative coefficients maintain statistical significance at conventional levels across all buffer specifications from 5km to 50km. This stability validates our 20km baseline choice and indicates that our findings do not depend on a particular geographic boundary selection.

Controlling for Nighttime Light Density as a Proxy for Regional Development

A central identification concern is that 3G network expansion may coincide with broader economic development, potentially confounding our estimates. Following common practice in the development and urban economics literature that uses satellite-derived nighttime lights as a proxy for local economic activity (Adema, Aksoy and Poutvaara, 2022), we augment the baseline specification with a control for nighttime light (NTL) density. We construct baseline NTL density using the Annual VIIRS Nighttime Lights (DNB) v2.1 "average (masked)" radiance product (Goodman et al., 2019)[5], computing area-weighted mean radiance within each ADM2 polygon (LGA) in 2012 (the pre-sample baseline year). We then interact this baseline NTL measure with year fixed effects, yielding a set of LGA-specific luminosity trends that flexibly absorb differential economic trajectories across areas without introducing a time-varying control that may itself be affected by 3G expansion. This approach avoids the bad-control bias that would arise from conditioning on contemporaneous or lagged NTL values that post-date the onset of treatment.

Table A2 reports results with the baseline NTL $\times$ Year FE control across our main fixed effects specifications. The inclusion of this control does not materially alter the estimated effects of 3G coverage: coefficients remain stable in sign and magnitude relative to the baseline. For women aged 12–20, the 3G coefficients remain -0.015^{***} to -0.020^{***} across specifications (compared to -0.014^{***} to -0.019^{***} in the baseline), indicating that our findings are not merely capturing general regional development trends. Conditioning on baseline luminosity trajectories thus yields estimates that are robust to differences in pre-existing economic intensity across local governments.

Technology-Specific Falsification Tests

We use the expansion of 2G networks as a falsification test to assess whether our findings are driven by general improvements in mobile communication rather than internet-specific capabilities. Unlike 3G, which enables internet browsing, 2G technology is limited to voice calls and SMS. If the mechanism we propose operates through internet access, 2G expansion should not generate fertility effects comparable to those of 3G. To test this, we re-estimate

[5] Dataset was accessed from AidData GeoQuery

our specifications replacing 3G with 2G coverage (constructed analogously), and also include both measures jointly. Table A2 reports the results. Across age groups, 2G coverage does not exhibit significant negative effects for the full sample; conversely, for women aged 12–20, 2G coefficients are significantly positive ($+0.012^{***}$ across all specifications), potentially indicating that basic mobile communication channels facilitate social connections that encourage earlier fertility rather than providing access to information that delays reproductive decisions. Crucially, when both 2G and 3G coverage are included simultaneously, the 3G coefficients for adolescents remain negative and statistically significant (-0.011^{***} to -0.017^{***}), while the 2G effects remain positive ($+0.009^{**}$ to $+0.010^{***}$). This divergence in sign reinforces our interpretation that it is the internet-enabled features of mobile networks—rather than improvements in voice and SMS communication—that drive the fertility responses we document. The consistency of our findings across these diverse robustness checks strengthens confidence in the causal interpretation of our estimates and the validity of our identification strategy.

6 Fertility Delay and Human Capital Reallocation

Economic models of fertility demonstrate that women's education and wage opportunities create substantial opportunity costs for childbearing, leading to the quantity-quality tradeoff where parents invest more in children's human capital as their own economic opportunities expand (Becker, 1960; Galor and Weil, 1993). Understanding how mobile internet access influences reproductive timing requires examining the fundamental life course transitions that precede fertility decisions—first cohabitation and first birth represent irreversible transitions that fundamentally alter women's opportunity sets and time allocation constraints, making the pre-transition period critical for understanding how technology shapes life trajectories. This creates what we term the "missing activity puzzle": if mobile internet enables young women to delay family formation by expanding their perception of alternative life opportunities, economic theory predicts that this freed time must be reallocated toward other productive activities. The primary candidate for this reallocation is human capital accumulation through continued education, as educational investment generates sustained productivity growth and raises future earning potential. Having established that mobile internet reduces fertility specifically among young women aged 12–20, we now investigate this reallocation mechanism by examining how digital connectivity affects educational attainment during the critical period when women delay major life transitions.

First Cohabitation and First Birth

If mobile internet enables young women to optimize reproductive timing through expanded economic opportunities, we should observe systematic delays in partnership formation and first births—key precursors to subsequent fertility decisions. This hypothesis follows from economic theory showing that when women perceive higher returns to delaying childbearing through career advancement or improved marriage prospects, they systematically postpone major life transitions (Rosenzweig and Wolpin, 1980; van Wijk and Billari, 2024).

We use the specification in Equation (1) as our identification strategy, replacing the outcome variable with an indicator for whether an individual begins cohabiting or gives birth in a specific year t. Since we are interested in the timing of first marriage or first birth, women who have already experienced these events are *right-censored*. Following Corno, Hildebrandt and Voena (2020), each woman contributes one observation for every at-risk year until the event occurs, and then *exits the sample*. This approach is consistent with duration models where the individual is only considered "at risk" of experiencing the event until it first happens. Once a woman gets married or gives birth, she is no longer exposed to the risk of a new first marriage or first birth; keeping her in the panel would artificially deflate the hazard rate by including periods where the event cannot occur.

For example, a woman who married at age 16 would appear five times in the regression for child marriage, starting from 2013 when she was age 12. Her marriage vector would be:

$$\{M_{ic,13}, \ldots, M_{ic,16}\} = \{0, \ldots, 0, 1\}.$$

In contrast, a woman who remains unmarried by the survey year contributes a sequence of zeros, as she is still at risk of first marriage but has not yet experienced the event. This right-censoring setup aligns with the discrete-time hazard model used in Corno, Hildebrandt and Voena (2020), ensuring that the analysis correctly represents exposure to the event of interest and prevents double-counting individuals beyond their relevant risk period.

Table 4 present estimates of mobile internet coverage effects on first cohabitation and first birth outcomes, focusing on the full sample and the age group (12-20) that showed the strongest fertility responses. The results demonstrate consistent negative treatment effects across both life course transitions that tell a coherent story about delayed family formation.

For first cohabitation, the full sample analysis shows statistically significant negative coefficients where a one standard deviation increase in 3G coverage reduces cohabitation probability by 1.8-3.2 percentage points across specifications. The effects become larger and more significant with more stringent fixed effects, suggesting that the results are not driven by unobserved confounders. The baseline cohabitation rate of 9.0% means these represent substantial 20-35.6% reductions relative to the control group. The age-restricted analysis (Panel B) reveals even stronger effects among women aged 12-20, with cohabitation probability reductions ranging from 3.0-3.9 percentage points. Given the lower baseline rate of 8.5% for this younger group, these effects represent a remarkable 35.3-45.9% reductions in cohabitation probability, indicating that mobile internet access fundamentally alters young women's partnership timing decisions.

Similarly, for first birth outcomes, the effects follow the same pattern with consistent negative treatment effects. The full sample shows 1.8-3.4 percentage point reductions in first birth probability, representing 24.3-45.9% decreases relative to the 7.4% baseline rate.

Among women aged 12-20, the effects range from 2.8-3.8 percentage points, translating to 43-58.5% reductions relative to their 6.5% baseline first birth rate. The systematic delays in both first cohabitation and first childbearing provide compelling evidence that mobile internet access enables young women to postpone major life transitions. This finding is crucial because it demonstrates that the technology operates by expanding perceived life opportunities and altering preferences about optimal timing for family formation.

Suggestive Evidence on Educational Attainment: Human Capital Investment During Delayed Family Formation

Nigeria operates under the Universal Basic Education (UBE) scheme following a 6-3-3 structure: primary education spans ages 6–12 (6 years), junior secondary covers ages 12–15 (3 years), and senior secondary extends from ages 15–18 (3 years), with official completion at 12 total years of schooling. This institutional framework provides critical context for interpreting educational outcomes, as our target population aged 12–25 encompasses women at key educational transition points—from primary to junior secondary (age 12), junior to senior secondary (age 15), and secondary completion (age 18). The timing of 3G expansion during this critical period creates an opportunity to examine whether mobile internet access influences educational persistence and completion at these crucial junctures. Importantly, our fertility results indicate that the largest behavioral responses to mobile internet occur among adolescents and young women. Therefore, we expect educational effects to be concentrated among cohorts who were still making schooling decisions when 3G networks were rolled out, rather than among women who had already completed their education prior to 2012.

Our analytical approach divides the sample based on age-appropriate educational milestones to avoid misclassification bias. For secondary and high school completion analysis, we exclude women younger than the official start age since coding them as zero would in-correctly classify women who simply had not yet reached the relevant educational stage. We incorporate a two-year buffer beyond official completion ages to account for late school entry—for example, while women are officially expected to finish junior secondary by age 14, we include women who were aged 16 in 2012 when 3G rollout began. Our age calculations use survey year ages and track back to determine each woman's school age in 2012 during the initial phase of 3G expansion. For example, women aged 21–25 in the 2018 DHS were ages 15–19 when 3G expansion began in 2012 and were therefore at critical secondary-school transition ages. In contrast, women aged 15–19 in the 2018 DHS were ages 9–13 in 2012 and were primarily exposed during primary-school years. For the educational gap measure, we restrict the sample to cohorts who were of school age (6–18 years) during the rollout period; women older than 18 in 2012—who should have officially completed schooling prior to 3G expansion—are excluded from this analysis.

Given the cross-sectional nature of the DHS data, our empirical specification employs Local Government Area (LGA) fixed effects to control for time-invariant local infrastructure and economic conditions. Since mobile internet expansion in Nigeria began after 2012, we construct our key explanatory variable as the average 3G coverage between 2012 and the survey year to capture cumulative exposure to digital infrastructure. The estimation equation is specified as follows:

$$Education_{iclt} = \beta \times \overline{Coverage}_{ct} + \mathbf{X}'_{ict}\gamma + \mathbf{W}'_{ct-1}\psi + \mu_l + \delta_t + \varepsilon_{iclt}, \tag{4}$$

where $Education_{iclt}$ represents binary indicators for age-appropriate educational completion for woman i in cluster c, Local Government Area l, at time t. Outcome variables include primary completion for younger cohorts (ages 15–19), secondary completion for intermediate cohorts (ages 18–22), and high school completion for older cohorts (ages 21–25), as well as a continuous measure of the educational gap defined as years behind the official 12-year completion standard for cohorts exposed to 3G during school age. The explanatory variable $\overline{Coverage}_{ct}$ denotes standardized average 3G coverage, proxying cumulative digital infrastructure exposure. The vector $\mathbf{X}_{ict}$ includes individual and household characteristics measured at the survey year—specifically age and household wealth index, while $\mathbf{W}_{ct-1}$ captures lagged weather conditions, including precipitation, temperature, solar radiation, wind speed, vapour pressure, and rainfall. All specifications include LGA fixed effects μ_l and survey year fixed effects δ_t.

Table 5 documents positive and statistically significant effects of 3G coverage across multiple cohorts and educational milestones. Primary completion increases by 7.0 percentage points for women aged 15–19 ($p < 0.05$), 9.2 percentage points for women aged 18–22 ($p < 0.01$), and 10.2 percentage points for women aged 21–25 ($p < 0.01$), indicating broad-based gains in foundational schooling. Effects are especially large at later educational transitions: secondary completion rises by 11.7 percentage points for women aged 18–22 (relative to a 28.1% baseline) and 16.0 percentage points for women aged 21–25 (relative to a 23.9% baseline). High school completion increases by 13.1 percentage points for the oldest cohort, a large gain relative to the low baseline completion rate of 3.2%. The educational gap measure confirms cumulative gains: for cohorts who were of school age during the rollout period, the gap behind the official 12-year completion benchmark narrows by 1.1 years from a baseline of 7.9 years.

Importantly, these patterns closely mirror our fertility results. The strongest educational gains occur among cohorts who were adolescents during the rollout period, the same population for which we observe substantial delays in cohabitation and childbearing. This consistency suggests that part of the fertility response operates through increased human capital investment, as mobile internet access encourages young women to remain in school during the years when marriage and fertility decisions are most salient.

These educational gains align with human capital theory's predictions regarding optimal investment timing and opportunity cost adjustments. Mobile internet access raises the perceived returns to education by expanding access to information on skilled occupations and credential-dependent career paths, while lowering the costs of continued schooling through digital learning resources. The concentration of effects at secondary and high school levels suggests that technology access enables women to remain in school during periods when they might otherwise exit for marriage or family formation, shifting the opportunity cost calculus in favor of continued human capital investment. This educational upgrading complements our fertility results by reinforcing incentives for delayed childbearing, as higher educational attainment increases future earnings potential and raises the opportunity cost of early family formation. Taken together, the education and fertility findings suggest that mobile internet primarily affects life-course decisions among young women whose educational and family formation trajectories were still flexible when 3G technology became available.

7 Suggestive Evidence: Contraceptive Adoption

Beyond the direct effects on time allocation, mobile internet may also influence fertility through a supply-side channel operating within existing partnerships: improved knowledge and adoption of contraceptive methods. This mechanism is particularly relevant for married and cohabiting women, whose partnerships are already established, and it is important to investigate because it helps distinguish between demand-side explanations (where women actively delay fertility due to expanded economic opportunities) and supply-side explanations (where women desire smaller families but lack the means to achieve their preferred outcomes). If mobile internet affects fertility primarily through information and reproductive health literacy, we would expect to see significant increases in contraceptive adoption among young women who simultaneously exhibit the strongest fertility responses.

Contraception Usage History

While contraceptive access is widely recognized as fundamental to fertility control, recent experimental evidence reveals mixed results regarding its effectiveness in high-fertility settings. Dupas et al. (2024) provided free contraception for three years in a large-scale randomized trial covering 14,545 households in rural Burkina Faso, finding precisely zero effect on fertility despite 20% higher voucher usage, concluding that fertility levels are primarily determined by deep economic factors rather than contraceptive availability. Conversely, Jensen (2012) demonstrates that when garment factories opened in rural India, providing economic opportunities for young women, fertility declined substantially through delayed marriage and economic empowerment rather than improved contraceptive access. Given that our results in Table 4 indicate fertility reductions stem mainly from delayed partnership formation, we expect mobile internet to affect single and married women through different pathways: among single women, fertility should fall via postponed cohabitation, while among married women—whose partnerships are already established—any fertility effects should operate through increased contraceptive uptake within existing relationships. Accordingly, we test whether mobile internet increases contraceptive use among women who have already formed partnerships to distinguish between economic empowerment and contraceptive access mechanisms.

While traditional contraception methods are often imprecise with high adherence costs, modern methods require awareness and social acceptability; since mobile internet disseminates contraceptive information and reduces knowledge barriers, we expect stronger effects on modern contraceptive adoption. The DHS recorded contraception use in the past 80 months since the survey year. Based on this information, we group contraceptive use into three variables: Any method includes use of any contraceptive method; modern methods includes female sterilization, male sterilization, contraceptive pills, IUDs, injectables, implants, female condoms, male condoms, emergency contraception, lactational amenorrhea method

(LAM), standard days method (SDM), and other country-specific modern methods but excludes abortions and menstrual regulation; traditional methods includes periodic abstinence, withdrawal, and country-specific traditional or folk methods of unproven effectiveness such as herbs, amulets, and spiritual methods.

We use Equation (1) but replace outcomes with contraception usage. The variables get at the average frequency with which the respondent practices any method, modern methods or traditional methods respectively in a given year. In a specific year t, we look at the respondent's family planning strategy from last birth to the next conception or, if sooner, calendar date at which the interview is conducted. We start counting usage from 6 months after birth to account for the time it takes to return to normal reproductive functions, especially if mothers are breastfeeding. Based on this we calculate the proportion of months (scaled between 0-1) during which the respondent used any family planning method, modern method or a transitional method in a given year t.

The contraception analysis in Table 6 reveals a striking pattern that challenges conventional assumptions about how digital connectivity affects reproductive behavior. For the full sample (Panel A), modern contraception shows consistently positive and significant effects of 1.1–1.3 percentage points across all specifications ($p < 0.01$). Any contraception use rises by 1.7 percentage points in the baseline specification ($p < 0.01$), though this effect attenuates and loses significance once state-by-year fixed effects are included, suggesting some confounding by state-level public health programs. These results indicate that mobile internet access raises adoption of modern contraceptive methods such as pills, IUDs, and injectables among the general population, likely through improved reproductive health literacy and awareness of family planning services. However, Panel B reveals a crucial contrast: among women aged 12–20—the exact demographic showing the strongest fertility reductions—mobile internet has no significant effects on contraceptive use across any method or specification. All coefficients are small in magnitude and statistically insignificant. This null result creates an apparent paradox when considered alongside the strong negative fertility effects documented for the same age group, and fundamentally challenges the view that fertility reductions in response to mobile internet must operate through improved contraceptive access.

To address potential recall bias concerns, we conducted robustness checks limiting the sample to three years of contraceptive history for the 2018 DHS cohort. The results in Table A3 remain consistent: modern contraception use is significantly higher in the full sample (0.011** in the preferred specification), while effects on women aged 12–20 remain small and insignificant across all methods and specifications. Using a short panel of all women regardless of marital status yields the same qualitative pattern (Table A4).

The absence of contraceptive effects among young women despite their strong fertility responses likely reflects several constraining mechanisms. Given that a large share of respondents obtain contraceptives from the public sector, internet access alone has limited power to shift contraceptive choices when supply-side barriers persist. More fundamentally, the null contraceptive results suggest that the fertility reductions we document among women aged 12–20 operate primarily through delayed family formation—postponing first cohabitation and first birth—rather than through increased contraceptive use within existing partnerships. These findings indicate that enhanced information access raises aspirations and expands perceived outside options, but structural barriers to reproductive autonomy within partnerships limit the scope for information alone to shift contraceptive behavior among adolescents.

8 Heterogeneous Effects

While the preceding average effects reveal clear age patterns, understanding the sources of heterogeneity within age groups sheds further light on the mechanisms through which mobile internet shapes fertility. We examine two dimensions: the role of women's pre-existing reproductive history in moderating fertility responses, and whether effects differ systematically across geographic and demographic contexts within Nigeria.

Reproductive History and Fertility Responses

Economic models of fertility demonstrate that women's decisions about additional children depend critically on the number and survival of children already born (Becker, 1960; Caldwell, 1980). Children traditionally serve as economic assets in developing contexts, providing labor, old-age security, and social status, creating preferences for larger family sizes. At the same time, families with many existing children face escalating physical and financial costs of additional births, creating strong incentives to cease childbearing. The number of children already born thus becomes a critical determinant of marginal utility calculations for subsequent births.

Child mortality experience adds another layer of complexity to these decisions. High mortality rates may encourage couples to have additional children as insurance against future losses, following a "hoarding" strategy (Cohen and Montgomery, 1988; Ben-Porath, 1976). Conversely, families who have experienced child deaths may become more aware of the substantial emotional and financial costs associated with repeated childbearing and loss, potentially reducing desired fertility through the quantity-quality trade-off formalized by Becker and Lewis (1973).

We examine interactions between 3G coverage and women's reproductive history prior to the technology rollout to capture how accumulated childbearing shapes responses to mobile internet. Table 7 reveals a fundamental asymmetry that helps explain the age-patterned results. For women aged 12–20 (Panel B), mobile internet coverage reduces birth probability by 2.4–2.9 percentage points ($p < 0.01$) for nulliparous women, but this effect is completely reversed for those with existing children: each additional prior birth increases birth probability by 5.2–5.3 percentage points ($p < 0.01$) in response to 3G coverage. This means young mothers respond to mobile internet by increasing subsequent fertility. This pattern is consistent with Becker's framework: young, childless women exposed to economic opportunities face high opportunity costs of early family formation and delay childbearing, while young mothers have already made the transition to motherhood, fundamentally altering their opportunity costs and life trajectory. For child mortality, interactions among young women are positive but statistically insignificant, reflecting the rarity of child deaths in this age group. For older women (age > 25, Panel D), the pattern reverses. A one standard deviation increase in 3G coverage increases birth probability by 1.4–1.7 percentage points ($p < 0.01$) for women without many children, but this positive effect is offset for those with larger existing families by approximately 0.3–0.4 percentage points per additional prior birth ($p < 0.01$). This suggests that among older women digital access enables more informed family planning decisions that reflect existing family size.

The age-stratified interaction patterns support the cohabitation delay interpretation of the main results. Effects concentrate among ages 12–20, where most women are nulliparous and marriage decisions remain flexible. The null effects among women aged 20–25 are consistent with this group containing mixed reproductive histories, where opposing effects cancel out statistically. For women over 25, mobile internet enables more strategic family planning among those with few children while moderating further fertility among those who have approached their desired family size. Importantly, the absence of significant child mortality interactions across all age groups suggests the fertility responses operate primarily through information and opportunity cost channels rather than through insurance motives.

Regional Heterogeneity

Nigeria exhibits large geographic variation in fertility rates, female education, and cultural norms around marriage and childbearing, particularly between the predominantly Muslim northern states and the more economically developed southern regions (Caldwell, 1980). These contextual differences raise the question of whether mobile internet's fertility effects are concentrated in the high-fertility north or operate similarly across the country.

Table 8 reports interaction models testing for differential effects by region (northern versus non-northern Nigeria) across all four age subgroups and three fixed effects specifications. The main effect captures the estimated effect for non-northern women; the interaction term captures the additional effect for northern women.

For women aged 12–20 (Panel B), the main 3G effect for non-northern women is -0.013^{***} across all three specifications. The northern interaction is -0.012^{**} in the preferred specifications ($p < 0.05$), indicating that northern adolescents experience fertility reductions roughly twice as large as those in the south. This amplification is consistent with northern Nigeria's higher baseline fertility rates, lower female education, and stronger patriarchal norms around early marriage—precisely the contexts where mobile internet's role in expanding aspirations and outside options is most likely to shift behavior at the margin.

For women aged 20–25 (Panel C), a different pattern emerges. Non-northern women in prime reproductive years show significant fertility reductions (-0.025^{**}, $p < 0.05$), but the northern interaction is significantly positive ($+0.044^{**}$, $p < 0.05$), largely offsetting the baseline effect for northern women in this age group. This divergence likely reflects compositional differences: northern women aged 20–25 are more likely to be married with young children, a stage at which mobile internet may primarily facilitate household coordination or improved maternal health information rather than delayed fertility. For women over 25 and for the full sample, regional interactions are small and statistically insignificant.

Taken together, the regional heterogeneity results show that the fertility-reducing effects of mobile internet among adolescents are stronger in the north, where information barriers to alternative life trajectories are highest, rather than concentrated in the more economically developed south. The reversal among northern women in prime reproductive age underscores that mobile internet's effects depend on the life stage and institutional context in which coverage is adopted.

9 Conclusion

We provide quasi-experimental estimates of the impact of mobile internet access on fertility by exploiting a natural experiment created by the staggered rollout of 3G networks across Nigeria between 2013 and 2018. Combining high-resolution georeferenced 3G coverage data with the 2018 Nigerian Demographic and Health Survey—from which we reconstruct complete birth histories—in a two-way fixed effects framework complemented by the Callaway-Sant'Anna heterogeneity-robust staggered difference-in-differences estimator, we find that mobile internet expansion significantly reduced adolescent fertility. Specifically, a one standard deviation increase in local 3G coverage lowers the annual probability of birth among women aged 12–20 by 1.4–1.9 percentage points, corresponding to a 12–17 percent decline relative to baseline fertility.

Mechanism analyses indicate that these fertility reductions operate through economic opportunity channels rather than traditional family-planning pathways. Mobile internet exposure significantly improves educational attainment among exposed cohorts: secondary completion rises by 11–16 percentage points and the gap behind the official 12-year completion benchmark narrows by 1.1 years, with effects concentrated among cohorts who were adolescents when 3G networks rolled out. In contrast, we find no evidence that mobile internet exposure increases contraceptive adoption among young women; contraceptive use remains unchanged across all methods. The fertility response is driven entirely by delayed partnership formation and postponed age at first birth, consistent with a framework in which technology raises the opportunity cost of early childbearing by expanding educational aspirations. We further document heterogeneity by reproductive history and region. Among nulliparous young women, mobile internet substantially delays entry into motherhood; women who had already begun childbearing prior to network rollout experience higher subsequent fertility, consistent with standard life-cycle models in which the opportunity cost of additional births varies with prior fertility. Effects are amplified in northern Nigeria—where baseline fertility, early marriage rates, and barriers to education are highest—and attenuated or reversed among northern women in prime reproductive years, reflecting compositional differences in partnership and childbearing status across the country's regional contexts.

Several caveats are worth noting. Our analysis cannot speak directly to longer-term outcomes beyond the observation window, including completed fertility or lifetime economic trajectories. In addition, although self-reported fertility measures are standard in demographic research, they may be subject to recall or reporting bias. Finally, while the staggered rollout of mobile networks provides plausibly exogenous variation in connectivity, we cannot rule out all confounding factors, though extensive robustness checks mitigate these concerns. Despite these limitations, our findings contribute to understanding the microeconomic foundations of innovation-driven growth. Consistent with the frameworks of (Mokyr, 2018; Aghion and Howitt, 1992; Howitt, 1999), we show that telecommunications infrastructure operates as a growth-enhancing technology by displacing traditional information channels that constrain women's educational aspirations. By inducing women to delay childbearing during critical periods of human capital accumulation, mobile internet addresses the quality–quantity tradeoff central to endogenous growth theory and identifies a previously underexplored pathway through which information technology accelerates demographic transitions in developing economies.

The policy implications are straightforward. Investments in digital infrastructure may generate demographic dividends through female economic empowerment, complementing traditional reproductive health interventions. The concentration of effects among adolescents aged 12–20 highlights the importance of early exposure, while the amplification of effects in northern Nigeria—the country's highest-fertility region—suggests that equitable connectivity expansion may yield the largest returns precisely where fertility rates and education gaps are greatest. Ensuring broad connectivity is therefore crucial for realizing the full demographic and economic benefits of telecommunications expansion in Sub-Saharan Africa.

Future research should examine whether technology-induced fertility delays translate into lower completed fertility, explore the specific information channels through which mobile internet shapes women's expectations and choices, and assess the external validity of these mechanisms across institutional and cultural contexts.

Tables

Table 1: Descriptive Statistics (2018 DHS)

Variable	Mean	N
Demographics		
Age	29.157	41,623
Education (%)		
No education	34.5	41,623
Primary education	15.2	41,623
Secondary education	39.9	41,623
Higher education	10.4	41,623
Geographic and Birth		
Northern region (%)	37.0	41,623
Urban residence (%)	40.4	41,623
Islam (%)	50.2	41,623
Children under 5	1.342	41,623
Total children	3.051	41,623
Age at first birth	19.688	29,850
Age at first marriage	18.526	31,010
Contraception Knowledge and Use (%)		
Knows contraception	92.1	41,623
Uses contraception	13.5	41,623
Uses modern contraception	10.2	41,623
Contraception source	44.4	3,882
Employment (%)		
Currently employed	64.7	41,623
Employed past year	68.1	41,623
Husband employed	96.3	28,667

Notes: This table reports descriptive statistics for women in the 2018 Nigerian Demographic and Health Survey (DHS). Reported values are sample means unless otherwise indicated. Education categories refer to the highest level of schooling completed. Northern region indicates residence in one of Nigeria's northern states. Islam is an indicator equal to one if the respondent reports Muslim religious affiliation and zero otherwise, based on self-reported religion in the DHS. Employment variables capture current employment status and employment during the past 12 months. Contraceptive knowledge and use variables are defined following DHS standard definitions. Sample sizes vary across variables due to survey design and item nonresponse. Birth outcomes in the regression analyses are drawn from birth history records spanning 2013–2018 as reported by 2018 DHS respondents.

Table 2: Impact of Mobile Internet Coverage on Fertility and Temporal Placebo Tests

	(1)	(2)	(3)
Main Results: Birth Outcomes (2013-2018)			
Panel A: Full Sample			
3G Coverage (20km, t-1)	-0.001	-0.005**	-0.004*
	(0.002)	(0.002)	(0.002)
Observations	244,033	244,033	244,033
R-squared	0.208	0.209	0.216
Control Mean	0.192	0.192	0.192
Panel B: Age 12-20			
3G Coverage (20km, t-1)	-0.014***	-0.019***	-0.019***
	(0.003)	(0.003)	(0.003)
Observations	73,696	73,696	73,696
R-squared	0.291	0.294	0.297
Control Mean	0.114	0.114	0.114
Panel C: Age 20-25			
3G Coverage (20km, t-1)	0.007	-0.006	-0.006
	(0.006)	(0.009)	(0.009)
Observations	39,686	39,686	39,686
R-squared	0.221	0.226	0.226
Control Mean	0.289	0.289	0.289
Panel D: Age $>$ 25			
3G Coverage (20km, t-1)	0.004	0.002	0.003
	(0.003)	(0.003)	(0.003)
Observations	123,772	123,772	123,772
R-squared	0.208	0.210	0.211
Control Mean	0.204	0.204	0.204
Temporal Placebo Tests: Historical Birth Outcomes (2006-2011)			
Panel E: Full Sample (Placebo)			
3G Coverage (20km, t-1)	-0.006***	-0.003	-0.003
	(0.002)	(0.003)	(0.003)
Observations	233,082	233,082	233,082
R-squared	0.277	0.279	0.285
Control Mean	0.213	0.213	0.213
Panel F: Age 12-20 (Placebo)			
3G Coverage (20km, t-1)	-0.001	-0.000	-0.001
	(0.001)	(0.002)	(0.002)
Observations	36,828	36,828	36,828
R-squared	0.424	0.429	0.432
Control Mean	0.014	0.014	0.014
Individual FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State $\times$ Year FE	No	Yes	Yes
Age Cohort FE	No	No	Yes

Notes: Panels A-D report main results where dependent variable is a binary indicator for whether a woman gives birth in year t (2013-2018 period). Panels E-F report temporal placebo tests using historical birth outcomes from 2006-2011 period with 3G coverage measured in 2012-2017. The placebo test examines whether future network deployment predicts historical fertility patterns. 3G Coverage represents the standardized proportion of 3G coverage within 20km of survey cluster in year t-1. Control Mean represents the baseline birth rate when 3G coverage equals zero. All models include climate controls (precipitation, solar radiation, wind speed, vapor pressure, and temperature). Standard errors are clustered at the cluster level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Robustness Checks with Event Studies

	Callaway & Sant'Anna (2021)		Roth (2022) Power Analysis			
	ATT	Pre-trend Test	50%		80%	
	(SE)	χ^2, p-value	Slope	Bayes	Slope	Bayes
Panel A: Full Sample						
Estimate	-0.009	$\chi^2(4) = 1.28$, $p = 0.865$	0.005	0.565	0.007	0.226
	(0.007)					
95% CI	$[-0.024, 0.006]$					
Panel B: Age 12–20						
Estimate	-0.021^{***}	$\chi^2(4) = 2.02$, $p = 0.731$	0.006	0.564	0.009	0.226
	(0.008)					
95% CI	$[-0.037, -0.006]$					
Panel C: Age 21–25						
Estimate	-0.015	$\chi^2(4) = 9.64$, $p = 0.047$	0.016	0.568	0.024	0.227
	(0.025)					
95% CI	$[-0.064, 0.034]$					
Panel D: Age 25+						
Estimate	0.003	$\chi^2(4) = 1.84$, $p = 0.764$	0.007	0.566	0.011	0.228
	(0.011)					
95% CI	$[-0.019, 0.025]$					

Notes: This table presents difference-in-differences estimates using the Callaway and Sant'Anna (2021) estimator for settings with staggered treatment adoption. Columns 1–2 report the ATT (simple weighted average treatment effect) and the joint pre-trend test that all pre-treatment coefficients equal zero. Columns 3–6 report power analysis from Roth (2024) showing the linear trend slope detectable with 50% and 80% power and the corresponding Bayes Factors. Lower Bayes Factors indicate stronger evidence for parallel trends. Standard errors in parentheses, clustered at the cluster level. 95% confidence intervals reported below estimates.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 4: Impact of Mobile Internet Coverage on First Cohabitation and First Birth

	First Cohabitation			First Birth		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Full Sample						
3G Coverage (20km, t-1)	-0.018***	-0.026***	-0.032***	-0.018***	-0.026***	-0.034***
	(0.003)	(0.005)	(0.005)	(0.003)	(0.004)	(0.004)
Observations	77,126	77,126	77,126	86,843	86,843	86,843
R-squared	0.349	0.360	0.369	0.336	0.341	0.359
Control Mean	0.090	0.090	0.090	0.074	0.074	0.074
Panel B: Age 12–20						
3G Coverage (20km, t-1)	-0.030***	-0.037***	-0.039***	-0.028***	-0.035***	-0.038***
	(0.003)	(0.005)	(0.005)	(0.003)	(0.004)	(0.004)
Observations	55,884	55,884	55,884	62,370	62,370	62,370
R-squared	0.355	0.378	0.382	0.337	0.349	0.366
Control Mean	0.085	0.085	0.085	0.065	0.065	0.065
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	No	Yes	Yes	No	Yes	Yes
Age Cohort FE	No	No	Yes	No	No	Yes

Notes: Individuals exit the analysis in the year after their first cohabitation (columns 1–3) or first birth (columns 4–6). Dependent variables are binary indicators for whether a woman enters first cohabitation or gives first birth in year t. 3G Coverage represents the standardized proportion of 3G coverage within 20km of survey cluster in year $t-1$. Control Mean represents the baseline rate when 3G coverage equals zero. All models include weather controls (precipitation, solar radiation, wind speed, vapour pressure, temperature, rainfall). Standard errors are clustered at the cluster level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Impact of Mobile Internet on Educational Outcomes by Exposure Cohort

	Education Completion by Survey Age (2018 DHS)						Educational Gap
	15–19	18–22		21–25			
	(9–13 in 2012)	(12–16 in 2012)		(15–19 in 2012)			(6–18 in 2012)
	Primary	Primary	Secondary	Primary	Secondary	High School	Education Gap
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3G Coverage (20km, Average)	0.070**	0.092***	0.117***	0.102***	0.160***	0.131***	−1.113***
	(0.031)	(0.029)	(0.033)	(0.026)	(0.029)	(0.033)	(0.278)
Observations	8,380	7,729	7,729	7,121	7,121	7,121	15,200
Control Mean	0.419	0.390	0.281	0.338	0.239	0.032	7.904
LGA FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: All columns use the 2018 DHS. Column headings report respondents' age at the 2018 survey. Ages shown in parentheses indicate respondents' age in 2012, the beginning of Nigeria's 3G mobile internet expansion period. Nigeria follows a 6–3–3 education system, with primary school typically completed by age 12, junior secondary school by age 15, and senior secondary school by age 18. A two-year buffer is allowed to account for delayed school entry and grade repetition. Columns (1), (2), and (4) examine primary school completion; columns (3) and (5) examine secondary school completion; column (6) examines high school completion. Columns (1)–(6) focus on cohorts who were at age-appropriate stages of schooling during the 3G rollout period. Column (7) reports the educational gap, defined as the difference between expected and completed years of schooling among women who were school-age (ages 6–18) in 2012. 3G coverage is measured as the average coverage between 2012 and the 2018 survey year and standardized. Weather controls include precipitation, solar radiation, wind speed, vapour pressure, temperature, and rainfall. Standard errors are clustered at the survey cluster level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Impact of Mobile Internet on Contraceptive Use Among Cohabitated Women

	Any Contraception			Modern Contraception			Traditional Contraception		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Full Sample									
3G Coverage (20km, t-1)	0.017***	0.009	0.008	0.011***	0.013***	0.013***	0.005	-0.004	-0.004
	(0.004)	(0.006)	(0.006)	(0.003)	(0.005)	(0.005)	(0.003)	(0.003)	(0.003)
Observations	70,936	70,936	70,935	70,936	70,936	70,935	70,936	70,936	70,935
R-squared	0.723	0.726	0.727	0.709	0.711	0.712	0.742	0.744	0.745
Control Mean	0.046	0.046	0.046	0.032	0.032	0.032	0.014	0.014	0.014
Panel B: Age 12-20									
3G Coverage (20km, t-1)	0.011	0.008	0.008	0.008	0.013	0.012	0.003	-0.004	-0.004
	(0.013)	(0.016)	(0.016)	(0.011)	(0.014)	(0.014)	(0.008)	(0.007)	(0.007)
Observations	8,261	8,260	8,259	8,261	8,260	8,259	8,261	8,260	8,259
R-squared	0.721	0.732	0.732	0.668	0.680	0.680	0.820	0.832	0.832
Control Mean	0.026	0.026	0.026	0.019	0.019	0.019	0.007	0.007	0.007
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Age Cohort FE	No	No	Yes	No	No	Yes	No	No	Yes

Note: Sample includes all cohabitated women from the full dataset. Dependent variable is a binary indicator for contraceptive use in year t. Any Contraception includes both modern and traditional methods. Modern Contraception includes pills, IUDs, injectables, implants, condoms, and sterilization. Traditional Contraception includes rhythm, withdrawal, and folk methods. 3G Coverage represents the standardized proportion of 3G coverage within 20km of survey cluster in year t-1. All models include weather controls (precipitation, solar radiation, wind speed, vapour pressure, temperature, rainfall). Standard errors are clustered at the cluster level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7: Heterogeneous Effect of Mobile Internet Coverage on Fertility with Birth Number or Children Mortality before 3G Roll Out

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Full Sample						
3G Coverage (20km, t-1)	0.015***	0.012***	-0.002	-0.002	-0.009**	-0.002
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
3G × Total Births in 2011	-0.007***	-0.007***	-0.001*			
	(0.000)	(0.000)	(0.000)			
3G × Total Died Children in 2011				-0.020***	-0.018***	-0.004
				(0.005)	(0.005)	(0.005)
Observations	244,033	244,033	244,033	155,213	155,213	155,213
R-squared	0.209	0.210	0.216	0.112	0.114	0.129
Control Mean	0.192	0.192	0.192	0.266	0.266	0.266
Panel B: Age 12-20						
3G Coverage (20km, t-1)	-0.024***	-0.029***	-0.029***	0.012	-0.009	-0.000
	(0.002)	(0.003)	(0.003)	(0.013)	(0.015)	(0.015)
3G × Total Births in 2011	0.052***	0.053***	0.052***			
	(0.008)	(0.008)	(0.008)			
3G × Total Died Children in 2011				0.626	0.661	0.896
				(0.920)	(0.921)	(0.959)
Observations	73,696	73,696	73,696	24,907	24,907	24,907
R-squared	0.294	0.298	0.300	0.165	0.176	0.185
Control Mean	0.114	0.114	0.114	0.235	0.235	0.235
Panel C: Age 20-25						
3G Coverage (20km, t-1)	0.018***	0.004	0.003	0.013	-0.005	-0.005
	(0.006)	(0.009)	(0.009)	(0.009)	(0.012)	(0.012)
3G × Total Births in 2011	-0.009**	-0.007**	-0.007*			
	(0.004)	(0.004)	(0.004)			
3G × Total Died Children in 2011				-0.012	-0.006	-0.005
				(0.030)	(0.032)	(0.032)
Observations	39,686	39,686	39,686	31,522	31,522	31,522
R-squared	0.221	0.226	0.226	0.151	0.158	0.158
Control Mean	0.289	0.289	0.289	0.318	0.318	0.318
Panel D: Age > 25						
3G Coverage (20km, t-1)	0.016***	0.014***	0.017***	0.003	0.001	-0.000
	(0.003)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
3G × Total Births in 2011	-0.004***	-0.003***	-0.004***			
	(0.001)	(0.001)	(0.001)			
3G × Total Died Children in 2011				-0.005	-0.003	-0.002
				(0.005)	(0.005)	(0.005)
Observations	123,772	123,772	123,772	93,248	93,248	93,248
R-squared	0.208	0.210	0.211	0.153	0.156	0.156
Control Mean	0.204	0.204	0.204	0.254	0.254	0.254
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	No	Yes	Yes	No	Yes	Yes
Age Cohort FE	No	No	Yes	No	No	Yes

Note: Dependent variable is a binary indicator for whether a woman gives birth in year t. 3G Coverage is standardized proportion of coverage within 20km (t-1). The interaction terms capture how the effects of 3G coverage vary depending on pre-existing birth or death levels in 2011, prior to the 3G rollout. Control Mean is baseline when coverage=0. Weather controls included. Standard errors clustered at cluster level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8: Heterogeneous Effects by Region (Northern vs. Non-Northern Nigeria)

	(1)	(2)	(3)
Panel A: Full Sample			
3G Coverage (20km, t-1)	0.000	−0.006*	−0.005
	(0.002)	(0.003)	(0.003)
3G × Northern	−0.005	0.003	0.002
	(0.003)	(0.005)	(0.004)
Observations	244,033	244,033	244,033
R-squared	0.208	0.209	0.216
Control Mean	0.192	0.192	0.192
Panel B: Age 12–20			
3G Coverage (20km, t-1)	−0.013***	−0.013***	−0.013***
	(0.003)	(0.004)	(0.004)
3G × Northern	−0.004	−0.012**	−0.012**
	(0.005)	(0.006)	(0.006)
Observations	73,696	73,696	73,696
R-squared	0.291	0.295	0.297
Control Mean	0.114	0.114	0.114
Panel C: Age 20–25			
3G Coverage (20km, t-1)	0.008	−0.025**	−0.025**
	(0.007)	(0.011)	(0.011)
3G × Northern	−0.004	0.044**	0.044**
	(0.014)	(0.019)	(0.019)
Observations	39,686	39,686	39,686
R-squared	0.221	0.226	0.226
Control Mean	0.289	0.289	0.289
Panel D: Age > 25			
3G Coverage (20km, t-1)	0.005*	0.000	0.001
	(0.003)	(0.004)	(0.004)
3G × Northern	−0.006	0.005	0.005
	(0.005)	(0.007)	(0.007)
Observations	123,772	123,772	123,772
R-squared	0.208	0.210	0.211
Control Mean	0.204	0.204	0.204
Individual FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State × Year FE	No	Yes	Yes
Age Cohort FE	No	No	Yes

Note: Dependent variable is a binary indicator for whether a woman gives birth in year t. 3G Coverage is standardized 1-year lagged coverage within a 20km buffer. Northern is a binary indicator equal to one for clusters in Nigeria's northern geopolitical zones. The main effect is the estimated effect for non-northern women; the interaction captures the differential effect for northern women. Control Mean is the baseline birth rate when coverage = 0. Weather controls included. Standard errors clustered at cluster level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

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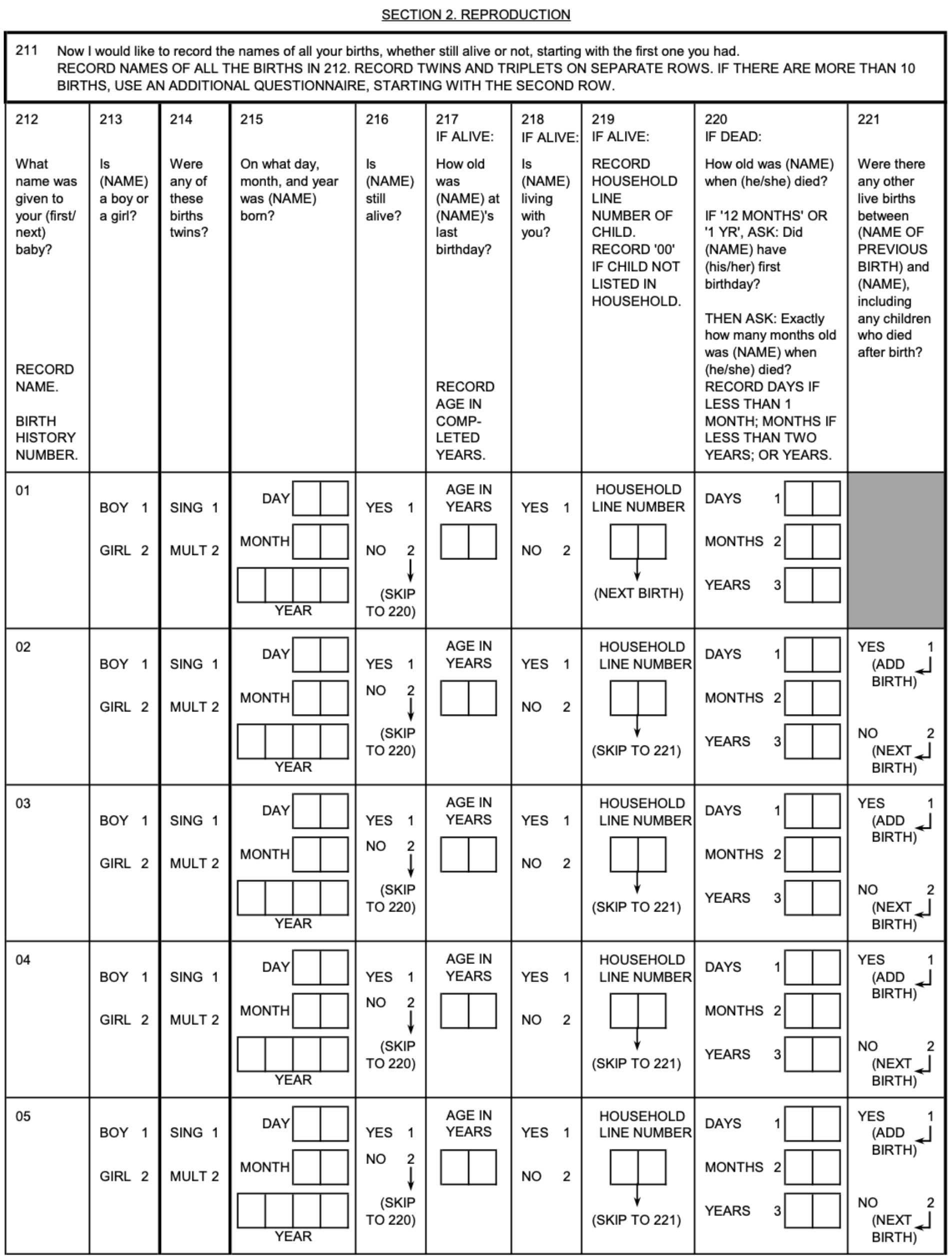
Appendix A1. Appendix Tables and Figures



SECTION 2. REPRODUCTION

211 Now I would like to record the names of all your births, whether still alive or not, starting with the first one you had.
RECORD NAMES OF ALL THE BIRTHS IN 212. RECORD TWINS AND TRIPLETS ON SEPARATE ROWS. IF THERE ARE MORE THAN 10 BIRTHS, USE AN ADDITIONAL QUESTIONNAIRE, STARTING WITH THE SECOND ROW.

212	213	214	215	216	217 IF ALIVE:	218 IF ALIVE:	219 IF ALIVE:	220 IF DEAD:	221
What name was given to your (first/ next) baby? RECORD NAME. BIRTH HISTORY NUMBER.	Is (NAME) a boy or a girl?	Were any of these births twins?	On what day, month, and year was (NAME) born?	Is (NAME) still alive?	How old was (NAME) at (NAME)'s last birthday? RECORD AGE IN COMP- LETED YEARS.	Is (NAME) living with you?	RECORD HOUSEHOLD LINE NUMBER OF CHILD. RECORD '00' IF CHILD NOT LISTED IN HOUSEHOLD.	How old was (NAME) when (he/she) died? IF '12 MONTHS' OR '1 YR', ASK: Did (NAME) have (his/her) first birthday? THEN ASK: Exactly how many months old was (NAME) when (he/she) died? RECORD DAYS IF LESS THAN 1 MONTH; MONTHS IF LESS THAN TWO YEARS; OR YEARS.	Were there any other live births between (NAME OF PREVIOUS BIRTH) and (NAME), including any children who died after birth?
01	BOY 1 GIRL 2	SING 1 MULT 2	DAY MONTH YEAR	YES 1 NO 2 (SKIP TO 220)	AGE IN YEARS	YES 1 NO 2	HOUSEHOLD LINE NUMBER (NEXT BIRTH)	DAYS 1 MONTHS 2 YEARS 3	
02	BOY 1 GIRL 2	SING 1 MULT 2	DAY MONTH YEAR	YES 1 NO 2 (SKIP TO 220)	AGE IN YEARS	YES 1 NO 2	HOUSEHOLD LINE NUMBER (SKIP TO 221)	DAYS 1 MONTHS 2 YEARS 3	YES 1 (ADD BIRTH) NO 2 (NEXT BIRTH)
03	BOY 1 GIRL 2	SING 1 MULT 2	DAY MONTH YEAR	YES 1 NO 2 (SKIP TO 220)	AGE IN YEARS	YES 1 NO 2	HOUSEHOLD LINE NUMBER (SKIP TO 221)	DAYS 1 MONTHS 2 YEARS 3	YES 1 (ADD BIRTH) NO 2 (NEXT BIRTH)
04	BOY 1 GIRL 2	SING 1 MULT 2	DAY MONTH YEAR	YES 1 NO 2 (SKIP TO 220)	AGE IN YEARS	YES 1 NO 2	HOUSEHOLD LINE NUMBER (SKIP TO 221)	DAYS 1 MONTHS 2 YEARS 3	YES 1 (ADD BIRTH) NO 2 (NEXT BIRTH)
05	BOY 1 GIRL 2	SING 1 MULT 2	DAY MONTH YEAR	YES 1 NO 2 (SKIP TO 220)	AGE IN YEARS	YES 1 NO 2	HOUSEHOLD LINE NUMBER (SKIP TO 221)	DAYS 1 MONTHS 2 YEARS 3	YES 1 (ADD BIRTH) NO 2 (NEXT BIRTH)

Figure A1: Birth Panel from DHS Survey

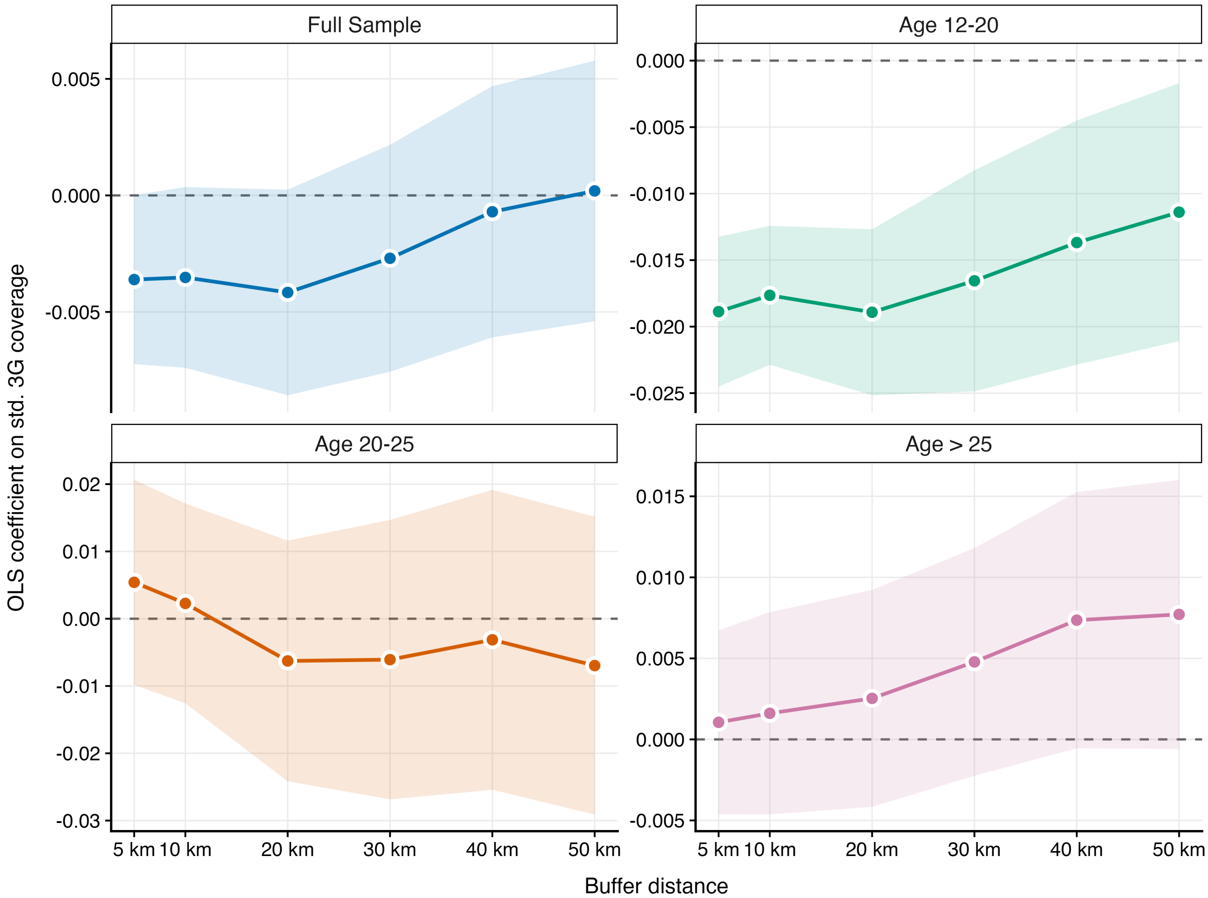


Figure A2: Effect of 3G Coverage on Fertility Across Buffer Distances (5–50 km) by Age Group

Note: Each panel plots OLS coefficient estimates on standardized 3G coverage (within the given buffer distance, lagged one year) for a separate age subgroup, with shaded 95% confidence interval ribbons. The dependent variable is a binary indicator for birth in year t. All specifications include individual, year, state×year, and age cohort fixed effects, plus climate controls (precipitation, solar radiation, wind speed, vapor pressure, temperature), with standard errors clustered at the survey cluster level.

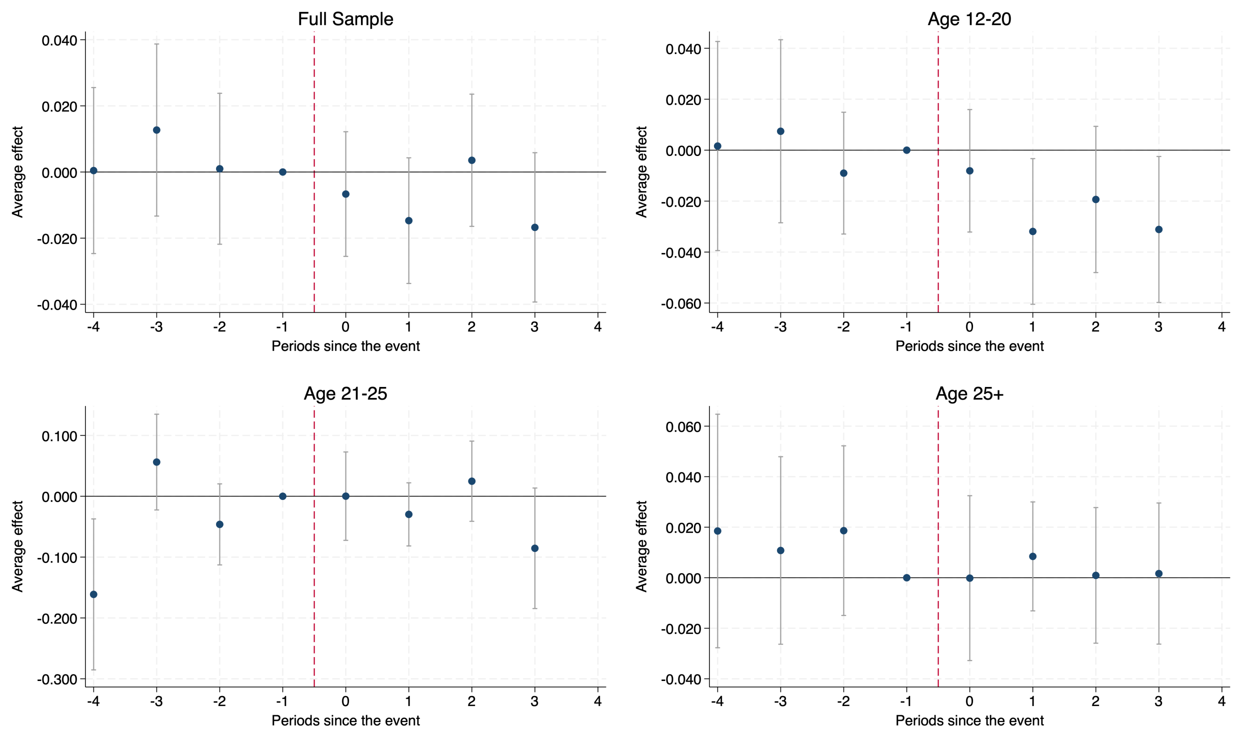


Figure A3: Callaway-Sant'Anna Event Study Estimates by Age Group

Note: This figure presents event-study estimates of treatment effects across four samples using the Callaway and Sant'Anna (2021) estimator with period -1 as the reference. Each panel plots coefficients for periods -4 to $+3$ relative to first 3G mobile coverage introduction (within 20km), with error bars representing 95% confidence intervals clustered at the primary sampling unit level. All specifications control for 2012 weather variables (precipitation and temperature), age, and use never-treated and not-yet-treated clusters as comparison groups. Pre-treatment coefficients (periods -4 to -2) test the parallel trends assumption. The Full Sample (Panel A), Age 12–20 (Panel B), and Age 25+ (Panel D) show pre-treatment estimates near zero with joint test p-values of 0.865, 0.731, and 0.764 respectively, providing strong support for parallel trends. Age 21–25 (Panel C) shows some pre-treatment variation ($p = 0.047$), though Roth (2022) power analysis indicates Bayes factors below 1 for all groups, suggesting the data remain more consistent with parallel trends than detectable violations. Post-treatment estimates reveal significant negative effects for Age 12–20 (ATT $= -0.021$, $p = 0.004$), with effects concentrated among younger women of reproductive age.

Appendix: Robustness Checks

Table A1: Double Machine Learning Robustness: Impact of 3G Coverage on Birth Outcomes

	Full Sample	Age 12–20	Age 21–25	Age > 25
LASSO	−0.003	−0.022***	−0.005	0.004
	(0.002)	(0.003)	(0.008)	(0.003)
ElasticNet	−0.004*	−0.021***	−0.006	0.004
	(0.002)	(0.003)	(0.008)	(0.003)
Random Forest	−0.005*	−0.018***	−0.015	0.002
	(0.003)	(0.004)	(0.010)	(0.003)
XGBoost	−0.002	−0.024***	0.004	0.003
	(0.002)	(0.003)	(0.008)	(0.003)
Observations	244,792	76,118	43,187	125,487

Notes: This table reports robustness checks using Double Machine Learning (DML) with a partially linear regression (PLR) specification. Each entry reports the average treatment effect for a one standard deviation increase in 3G coverage within 20km on the birth indicator. The DML procedure uses 5-fold cross-fitting (dml2) with four machine learning methods: LASSO (cv_glmnet, $\alpha = 1$), ElasticNet (cv_glmnet, $\alpha = 0.5$), Random Forest (ranger, 300 trees), and XGBoost (100 rounds). Controls include weather variables (precipitation, solar radiation, wind speed, vapour pressure, temperature) and state×year fixed effects (222 dummies). Individual fixed effects are approximated via within-group demeaning. Standard errors are clustered at the DHS cluster level. Sample: DHSYEAR=2018, GSMYEAR 2013–2018, age ≥ 12.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Mobile Internet Coverage and Fertility: Robustness and Falsification Tests

	3G + Nightlights			2G Falsification			2G + 3G Combined		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Full Sample									
3G Coverage (t-1)	-0.003*	-0.007***	-0.006**				-0.002	-0.005**	-0.004*
	(0.002)	(0.002)	(0.002)				(0.002)	(0.003)	(0.002)
2G Coverage (t-1)				0.000	0.000	0.001	-0.001	-0.001	-0.001
				(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.002)
Observations	244,033	244,033	244,033	244,033	244,033	244,033	244,033	244,033	244,033
R-squared	0.208	0.209	0.216	0.208	0.209	0.216	0.208	0.209	0.216
Control Mean	0.192	0.192	0.192	0.207	0.207	0.207	0.192	0.192	0.192
Panel B: Age 12–20									
3G Coverage (t-1)	-0.015***	-0.020***	-0.020***				-0.011***	-0.017***	-0.017***
	(0.003)	(0.003)	(0.003)				(0.003)	(0.003)	(0.003)
2G Coverage (t-1)				0.012***	0.012***	0.012***	0.010***	0.009**	0.009**
				(0.002)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Observations	73,696	73,696	73,696	73,696	73,696	73,696	73,696	73,696	73,696
R-squared	0.291	0.295	0.297	0.291	0.294	0.296	0.291	0.295	0.297
Control Mean	0.114	0.114	0.114	0.135	0.135	0.135	0.114	0.114	0.114
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Age Cohort FE	No	No	Yes	No	No	Yes	No	No	Yes
NTL $\times$ Year FE	Yes	Yes	Yes	No	No	No	No	No	No

Notes: Dependent variable is an indicator for birth in year t. Coverage variables are standardized within 20km radius, lagged one year. Columns (1)–(3) control for baseline (2012) nightlight density interacted with year fixed effects (NTL $\times$ Year FE), which absorbs differential luminosity trends across LGAs while avoiding bad-control bias from time-varying nightlights. Columns (4)–(6) present 2G falsification tests. Columns (7)–(9) include both 3G and 2G technologies. All specifications include climate controls. Standard errors clustered at survey cluster level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Impact of Mobile Internet on Contraceptive Use Among Cohabitated Women: Short Panel Results for 2015-2018 Cohort

	Any Contraception			Modern Contraception			Traditional Contraception		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Full Sample									
3G Coverage (20km, t-1)	0.011**	0.004	0.004	0.006*	0.011**	0.011**	0.005	-0.006*	-0.007*
	(0.005)	(0.006)	(0.006)	(0.004)	(0.005)	(0.005)	(0.003)	(0.004)	(0.004)
Observations	48,538	48,538	48,537	48,538	48,538	48,537	48,538	48,538	48,537
R-squared	0.806	0.807	0.807	0.811	0.812	0.812	0.802	0.803	0.803
Control Mean	0.046	0.046	0.046	0.033	0.033	0.033	0.013	0.013	0.013
Panel B: Age 12-20									
3G Coverage (20km, t-1)	0.015	0.007	0.007	0.009	0.010	0.010	0.007**	-0.003	-0.003
	(0.010)	(0.011)	(0.011)	(0.009)	(0.010)	(0.011)	(0.003)	(0.002)	(0.002)
Observations	4,993	4,993	4,992	4,993	4,993	4,992	4,993	4,993	4,992
R-squared	0.795	0.804	0.804	0.758	0.767	0.768	0.865	0.872	0.872
Control Mean	0.028	0.028	0.028	0.018	0.018	0.018	0.010	0.010	0.010
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Age Cohort FE	No	No	Yes	No	No	Yes	No	No	Yes

Note: Sample restricted to cohabitated women using the short panel (2015–2018) for the 2018 DHS cohort to address recall bias and ensure more accurate reporting of contraceptive use patterns. Dependent variable is a binary indicator for contraceptive use in year t. Any Contraception includes both modern and traditional methods. Modern Contraception includes pills, IUDs, injectables, implants, condoms, and sterilization. Traditional Contraception includes rhythm, withdrawal, and folk methods. 3G Coverage represents the standardized proportion of 3G coverage within 20km of survey cluster in year t-1. All models include weather controls (precipitation, solar radiation, wind speed, vapour pressure, temperature, rainfall). Standard errors are clustered at the cluster level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A4: Impact of Mobile Internet on Contraceptive Use Among Full Sample: Short Panel Results

	Any Contraception			Modern Contraception			Traditional Contraception		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Full Sample									
3G Coverage (20km, t-1)	0.011**	0.004	0.003	0.006	0.010**	0.010*	0.005	-0.006	-0.007
	(0.005)	(0.006)	(0.006)	(0.004)	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)
Observations	49,909	49,909	49,908	49,909	49,909	49,908	49,909	49,909	49,908
R-squared	0.807	0.808	0.808	0.811	0.813	0.813	0.800	0.801	0.802
Control Mean	0.047	0.047	0.047	0.034	0.034	0.034	0.013	0.013	0.013
Panel B: Age 12-20									
3G Coverage (20km, t-1)	0.018*	-0.012	-0.012	0.012	-0.003	-0.003	0.006*	-0.009	-0.008
	(0.010)	(0.012)	(0.013)	(0.009)	(0.011)	(0.011)	(0.003)	(0.006)	(0.006)
Observations	5,373	5,373	5,372	5,373	5,373	5,372	5,373	5,373	5,372
R-squared	0.801	0.811	0.812	0.769	0.778	0.779	0.864	0.876	0.876
Control Mean	0.029	0.029	0.029	0.019	0.019	0.019	0.010	0.010	0.010
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Age Cohort FE	No	No	Yes	No	No	Yes	No	No	Yes

Note: Sample includes all women (regardless of marital status) using the short panel (2015–2018) for the 2018 DHS cohort to address recall bias and ensure more accurate reporting of contraceptive use patterns. Dependent variable is a binary indicator for contraceptive use in year t. Any Contraception includes both modern and traditional methods. Modern Contraception includes pills, IUDs, injectables, implants, condoms, and sterilization. Traditional Contraception includes rhythm, withdrawal, and folk methods. 3G Coverage represents the standardized proportion of 3G coverage within 20km of survey cluster in year t-1. All models include weather controls (precipitation, solar radiation, wind speed, vapour pressure, temperature, rainfall). Standard errors are clustered at the cluster level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$